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Intangible Input Complementarities Along Supply Chains

Francesco Grigoli, Nicola Pierri, and Marvin Cardoza

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Prepared by Marvin Cardoza, Francesco Grigoli, and Nicola Pierri*

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ABSTRACT: We study whether buyer-supplier relationships feature complementarities in intangible inputs. Using administrative data from the Dominican Republic, we show that foreign demand routed through intangible-intensive buyers raises a supplier's own intangible purchase share, while demand from other buyers has no effect. Two event studies on firms not previously exposed to intangible-producing partners provide further evidence. Forming a first link with an intangible-industry buyer raises the probability of any intangible purchase by about 4 percentage points, and hiring a worker from an intangible-producing firm for the first time has an effect of the same size. A general equilibrium production network model with endogenous intangible input use, knowledge transfer through worker mobility, and endogenous intangible prices implies that supply-chain complementarities account for about 15 percent of aggregate intangible purchases and about 4 percent of output.

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WORKING PAPERS

Intangible Input Complementarities Along Supply Chains

Prepared by Marcin Cardoza, Francesco Grigoli, and Nicola Pierri[1]

[1] We thank colleagues at Georgetown University for discussions and comments on earlier drafts.

1 Introduction

Intangible inputs—such as software, R&D, consulting, data analytics, and organizational know-how—have emerged as the defining feature of modern production (Begenau *et al.*, 2018; Haskel and Westlake, 2018; Brynjolfsson *et al.*, 2021; Crouzet *et al.*, 2022; Babina *et al.*, 2024), yet a striking share of firms use little or none of such inputs.[1] Even in Belgium, one of the world's richest economies, 18.6 percent of firm-years record no intangible purchases at all (Bijnens *et al.*, 2025), and the scope for catch-up is even greater in developing economies. A more complete understanding of the drivers of intangible inputs' adoption is therefore of paramount importance, especially as AI becomes a main determinant of firm productivity.

This paper documents and quantifies the importance of complementarities in the use of intangible inputs along supply chains as a novel driver of intangible adoption. That is, the return to a firm's intangible inputs depends not only on the firm itself, but on the firms it sells to. A supplier that invests in automated order processing realizes no return on it while its principal buyer continues to place orders by telephone; a carrier that adopts real-time shipment tracking creates value only for those clients whose own systems can receive and act on the resulting data. A food producer that hires consultants to certify its quality processes recoups that cost only if its buyers—hotel chains or exporters with their own brand and compliance standards—reward certification in procurement. Where buyers are intangible-intensive, a supplier's own investment is worth more; where they are not, it may not be worth undertaking. Intangible adoption is in this sense a strategic complement along the supply chain, propagating through the input-output structure of the economy.[2]

We leverage administrative data from the Dominican Republic covering the period 2012–2024. Three features of the data make our analysis possible. First, the VAT registry records the universe of firm-to-firm transactions, allowing us to observe each firm's complete set of buyers and suppliers and to measure the intangible intensity of its trading partners. This type of firm-to-firm data has proven invaluable for studying production networks (Bernard *et al.*, 2019; Carvalho and Tahbaz-Salehi, 2019; Cardoza *et al.*, 2025), but has rarely been combined with measures of intangible inputs. Second, the VAT firm-to-firm transaction data identify each purchase by the supplier's industry, which we use to classify purchases of intangible services following the methodology of Bijnens *et al.* (2025), which, in turns, builds on the taxonomy of Corrado *et al.* (2005). Firm-level income tax records complement this with balance-sheet information—revenue, employment, and wage bill. Third, matched employer-employee data from the Social Security Treasury allow us to track workers across firms and identify when a firm hires a worker who previously worked at an intangible-producing firm.

[1] Throughout the paper, "intangible inputs" refers to purchases of intangible services—software and data processing, R&D and design, consulting, advertising, and training, following the Corrado *et al.* (2005) taxonomy as implemented by Bijnens *et al.* (2025)—observed in the VAT records. In-house intangible production, such as software developed by a firm's own staff, is not captured.

[2] Strategic complementarity is meant in the sense of Bulow *et al.* (1985), namely that a buyer's intangible purchases raise its supplier's marginal return to purchasing them as well.

We begin by documenting a set of facts. Intangible purchases account for 3.9 percent of total firm purchases in the Dominican Republic, and about 50 percent of firm-year observations report zero intangible expenditure. Firms whose buyers are more intangible-intensive purchase significantly more intangible services themselves, even after controlling for industry, firm size, and other observables. This correlation, while suggestive, does not establish causality, as firms with high intangible demand may select into relationships with intangible-intensive buyers, or common shocks may drive both. Using a Khwaja and Mian (2008) two-way fixed-effects design that absorbs every time-varying supplier and buyer characteristic, we find that a high-intangible supplier directs about 21 percent more trade value to a high-intangible buyer than other suppliers direct to that same buyer. This homophily shows that intangible-using firms sit in tightly clustered pockets of the network—the cross-sectional footprint one would expect if intangible input decisions reinforce one another along the supply chain. Establishing whether they do requires exogenous variation.

We rely on three different empirical exercises, each capturing a different aspect of the role supply chains play in intangible use. First, to capture a firm's response to its buyers' use of intangible inputs, we exploit that a firm's buyers are heterogeneous both in terms of their use of intangibles and in their exposure to demand shocks. If intangible inputs along the supply chain are complements, then we expect a supplier's intangible use to increase (decrease) when its more intangible-intensive buyers expand (contract). As productivity shocks could impact both a firm's intangible purchases and its buyers' size at the same time, we isolate variation in demand that originates entirely outside the Dominican economy. Following Dhyne *et al.* (2021), we build a firm-level export-demand shock from changes in world import demand for the HS six-digit products each exporter sells, and propagate it upstream through the domestic production network. For each supplier, we then split its buyer-side exposure into a component transmitted through buyers that already purchased intangibles before the shocks begin and a component transmitted through buyers that did not.

Local projections (Jordà and Taylor, 2025) show that demand routed through intangible-purchasing buyers raises the supplier's own intangible purchase share. Specifically, a demand shock that increases a firm's sales by 14 percent also increases its suppliers intangible purchase share by by 3.1 percentage points five years out if this buyer uses intangible inputs, whereas demand routed through other buyers has no impact at any horizon. This asymmetry is the hallmark of a supply-chain complementarity, suggesting that what matters is not how much demand a supplier receives, but who it comes from. Because these demand shocks travel through pre-existing relationships, whatever information already flows between the partners is held fixed; the design varies only the scale of demand moving through the link, which is what isolates the complementarity from learning. Reassuringly, we find no anticipation effect of the demand shocks on intangible purchases.

The same logic operates in reverse when demand collapses. We leverage the fact that our sam-

ple period covers the COVID-19 pandemic, which shut down tourism in the Dominican Republic.[3] We find that a supplier selling its entire output to intangible-using hotels and restaurants cut its intangible purchases by roughly 84 percent in 2020, an effect that vanishes by 2021—the transitory response the mechanism predicts of a temporary shock. Suppliers whose tourism buyers did not use intangibles show no such contraction, although the two responses are not statistically distinguishable from one another. These results are consistent with the notion of complementarities in intangible purchases along the supply chain, rather than simply learning about intangible inputs from commercial partners. This is because information, once transmitted, is not withdrawn when a buyer's demand collapses. Hence, the reversal in intangible purchases is consistent with a complementarity that prices current demand.

Second, we examine what the complementarity delivers when the network itself changes, which we capture by looking at the moment a supplier gains its first intangible-using buyer. Using a stacked event-study design (Cengiz *et al.*, 2019; Goodman-Bacon, 2021; Sun and Abraham, 2021) on suppliers that had never previously linked to an intangible-industry buyer, we find that forming the first such link raises the supplier's probability of any intangible purchase by about 4 percentage points on impact, growing to 5–6 percentage points over the following years.

Third, we turn to the role of the supply chain in driving intangible adoption by encouraging the diffusion of human capital. Workers' movements across firms are an important driver of knowledge diffusion (Stoyanov and Zubanov, 2012; Jarosch *et al.*, 2021). Moreover, a recent literature has highlighted that workers also carry "supply chain-specific" knowledge when they move (Cardoza *et al.*, 2025). In our setting, this implies that employees of intangible-producing firms may possess knowledge on how to use intangible inputs in the productions of other downstream services and goods. When workers move from intangible producing firms to new employers, their know-how could lower the cost of using intangible in production thus causing an increase in the likelihood of using such inputs. Using another stacked event-study design on firms that had never previously hired from an intangible-producing firm, we show that the first such hire raises the probability of any intangible purchase by about 4 percentage points on impact. The effect is immediate and persistent, stabilizing around 3–4 percentage points four years after the hire.

Our empirical strategies require the detailed and confidential firm-to-firm records available for the Dominican Republic, but the pattern they reveal has broader external validity. Merging country pair-level trade data with country-level intangible-investments data, we show that countries whose exports were directed toward intangible-intensive destinations before the sample period subsequently accumulated more intangible capital themselves, both among advanced economies and across a broader sample of countries. This finding suggests that intangible input complementarities may similarly be present also in other countries.

To quantify the aggregate importance of these mechanisms, we develop a general equilibrium model of a production network in the spirit of Acemoglu *et al.* (2012) and Baqaee and Farhi (2019),

[3] Tourism is a key sector of the Dominican economy, with 6.4 million non-resident air arrivals in 2019. The borders were closed in March 2020; arrivals fell to zero in the second quarter of 2020, and full-year arrivals came in 63 percent below their 2019 level, according to the Central Bank of the Dominican Republic.

augmented with three ingredients tailored to our setting. First, firms produce using a CES technology with labor, intermediates, and intangible inputs, where the effective return to intangibles depends on the intangible intensity of the firm's buyers—the supply-chain complementarity. Second, intangible services are produced by a specialized sector with endogenous prices, so that aggregate adoption waves bid up the price and partially dampen the complementarity effect. Third, a frictional labor market with search-and-matching (Mortensen and Pissarides, 1994; Petrongolo and Pissarides, 2001) allows workers to carry knowledge across firms. Workers who separate from intangible-producing firms reduce the fixed cost of intangible adoption at their new employer, in the spirit of the knowledge-transfer literature (Jarosch *et al.*, 2021; Lise and Postel-Vinay, 2020).

Underlying all three ingredients is a fixed cost of adoption (Comin and Hobijn, 2010), which generates the extensive margin we document in the data—a large mass of non-adopters alongside a tail of intangible-intensive firms. We calibrate the model to three moments. First, the semi-elasticity of supplier intangible purchases to demand routed through intangible-purchasing buyers, which pins down the complementarity parameter; second, the share of firms that purchase intangibles at all, which pins down the fixed cost of adoption; and third, the hiring event study, which pins down the knowledge-transfer parameter in partial equilibrium. The full general equilibrium counterfactual—in which all prices, wages, output, and the intangible service price adjust simultaneously—shows that supply-chain complementarities account for 15 percent of aggregate intangible purchases and 4.1 percent of aggregate output. In a partial equilibrium version of the model, complementarities account for 26 percent of aggregate intangible purchases, but in general equilibrium the price of intangible services acts as a shock absorber—lower demand makes intangibles cheaper, pulling some firms back in—and about 42 percent of the effect is offset.

These findings carry direct implications for technology diffusion. Because intangible input decisions are interdependent along the supply chain, uniform firm-level subsidies understate the return to targeted intervention. For instance, a digitalization program aimed at buyers that are central in the production network can set off adoption among their suppliers, a multiplier that policies treating firms in isolation would miss. The same interdependence, however, can sustain coordination failures in which no firm invests because its partners have not. Worker mobility offers a complementary lever. Because hiring from intangible-producing firms lowers the cost of adoption, policies that ease job transitions out of these sectors diffuse the capability to adopt alongside the incentive to do so. These considerations apply with particular force to AI and related general-purpose technologies, whose value depends on compatibility across trading partners.

Related literature Our paper contributes to several strands of the literature. First, we contribute to the literature on intangible capital and its role in economic growth (Corrado *et al.*, 2005, 2009; Haskel and Westlake, 2018; Crouzet *et al.*, 2022; De Ridder, 2024; Lashkari *et al.*, 2024). A key finding of this literature is that intangible investment is concentrated among large, productive firms and has become an increasingly important source of market power and productivity dispersion. While most of the existing work focuses on within-firm determinants of intangible investment—

such as complementarity with skilled labor, IT capital, or management practices (Bloom *et al.*, 2012; Brynjolfsson *et al.*, 2021)—we show that a firm's position in the production network is an important external determinant. The complementarity we identify implies that the concentration of intangible investment is partly a network phenomenon. Firms invest more when their partners do, creating clusters of intangible-intensive supply chains alongside clusters of low-technology firms.

Second, our paper is closely related to the literature on technology diffusion through supply chains. Halpern *et al.* (2015), Alfaro-Ureña *et al.* (2022), and Amiti *et al.* (2024) document that accessing foreign inputs and forming supply chain relationships with multinational or superstar firms raises firm performance. Recent work shows that the adoption of some technologies—such as robots or digital payments—diffuses across production networks (Bilgin *et al.*, 2026; Criscuolo *et al.*, 2025) and among transacting parties (Crouzet *et al.*, 2023; Alvarez *et al.*, 2023). Demir *et al.* (2024) study the interplay between IT infrastructure and supply chains. Closest in spirit to our mechanism, Antràs and Chor (2013) model complementarities between buyer and supplier investments, in their case shaping the organization of value chains. Compared to this literature, we focus on intangible inputs and isolate the complementarity from information diffusion through demand variation within pre-existing relationships, and document a hiring channel through which workers from intangible producers spur adoption.

Third, we contribute to the literature on production networks and shock propagation (Acemoglu *et al.*, 2012; Huneeus, 2018; Baqaee and Farhi, 2019; Korovkin *et al.*, 2026). This literature has established that microeconomic shocks propagate through input-output linkages with aggregate consequences, and that the structure of the network—in particular, the degree distribution and the concentration of central nodes—determines the magnitude of amplification. We extend this framework by showing that not all demand is alike. Demand shocks from intangible-intensive buyers generate qualitatively different supplier responses than demand from other buyers, because the former activate a complementarity channel that the latter do not. Our general equilibrium model quantifies the resulting network multiplier and shows that the endogenous price of intangible services partially offsets it.

Fourth, we contribute to the growing literature on knowledge diffusion through labor mobility. Jarosch *et al.* (2021) and Herkenhoff *et al.* (2024) develop structural models of learning from coworkers, showing that worker mobility transmits knowledge across firms with aggregate productivity effects; and Cardoza *et al.* (2025) document knowledge spillovers through worker mobility in production networks. Closest to our hiring design, Mion and Opromolla (2014) show that hiring managers with export experience leads firms to start exporting, and Tambe and Hitt (2014) document IT spillovers through IT-worker flows. We build on these insights by showing that worker mobility from intangible-producing firms specifically reduces the cost of intangible adoption at the destination firm, and we embed this mechanism in a structural model in which the knowledge distribution is pinned down by steady-state worker flows in a frictional labor market (Postel-Vinay and Robin, 2002; Lise and Postel-Vinay, 2020).

Outline The remainder of the paper is organized as follows. Section 2 describes the data and measurement. Section 3 presents descriptive evidence, including the intangible homophily in supply-chain matching. Section 4 presents the demand-shock decomposition, and section 5 two complementary event studies—forming new links with intangible-industry buyers and hiring workers from intangible-producing firms—each pairing an identification strategy with its results and robustness. Section 6 develops the structural model, calibrates it, and presents counterfactual exercises. Section 7 concludes.

2 Data

2.1 Data sources and sample

We combine multiple anonymized administrative datasets from the Dominican Republic. The first dataset, compiled by the Directorate General of Internal Taxes (*Dirección General de Impuestos Internos*, DGII), reports firm-level information for the universe of firms that file income taxes: assets, liabilities, revenue, expenditures, employment, and the wage bill. The second dataset includes firm-to-firm transactions from the VAT registry, which allow us to identify each firm's domestic buyers and suppliers. These data cover the period 2012–2024. The third dataset contains wage and demographic information on employees from the Social Security Treasury (*Tesorería de la Seguridad Social*, TSS).

The raw dataset covers 179,774 unique firms over the period 2012–2024, for a total of about 1.7 million firm-year observations. We apply sample restrictions in two layers. First, for descriptive purposes, we require firms to have at least two full-time employees. This restriction excludes single-person entities, which in some cases are created for tax reasons. This descriptive sample contains 83,953 unique firms and 569,615 firm-year observations. Second, for the regression analysis, we restrict the sample to firms with annual revenue of at least 10,000 Dominican pesos, a wage bill of at least 1,000 pesos, total VAT purchases of at least 1,000 pesos, and a firm-to-firm (B2B) purchase share of at least 10 percent of revenue.[4]

2.2 Measuring intangible purchases from VAT data

We construct a firm-level measure of intangible purchases following the methodology of Bijnens *et al.* (2025), who use business-to-business (B2B) transaction data to identify purchases of intangible services. The key idea is to classify supplier industries as intangible-producing based on the Corrado *et al.* (2005, 2009) taxonomy of intangible capital. We map their classification—originally in NACE Rev. 2 codes—to the Dominican Republic's CIIU system (equivalent to ISIC Rev. 3.1) using official UN correspondence tables.

Specifically, we classify each firm's purchases into three categories of intangible inputs:

[4]The last filter that every firm in the regression sample has meaningful exposure to the production network.

a. *Computerized information*: purchases from software development, IT consultancy, data processing, and database services (CIIU 7210–7292).

b. *Innovative property*: purchases from R&D services, entertainment and artistic originals, and design and engineering activities (CIIU 7310–7320, 7421–7422, 9211–9233).

c. *Economic competencies*: purchases from advertising, market research, management consulting, personnel provision, and education services (CIIU 7413–7430, 7414, 7491, 8030, 8090).

For each firm i in year t, we define the intangible purchase share as

$$s_{it}^{\text{intang}} = \frac{\sum_{j \in \mathcal{I}} \text{purchases}_{ijt}}{\sum_{j} \text{purchases}_{ijt}} \times 100, \tag{1}$$

where $\mathcal{I}$ denotes the set of intangible-producing suppliers.

To measure supply-chain partner intangible intensity, we compute the purchase-weighted average intangible share of each firm's suppliers and buyers:

$$\bar{s}_{it}^{\text{suppliers}} = \frac{\sum_{j} w_{ijt} \cdot s_{jt}^{\text{intang}}}{\sum_{j} w_{ijt}}, \tag{2}$$

where w_{ijt} is the value of purchases from supplier j, and s_{jt}^{intang} is supplier j's own intangible purchase share. We construct an analogous measure for the sales-weighted average intangible share of buyers.

3 Descriptive evidence

3.1 Intangible purchases

We first document the extent to which Dominican firms engage in intangible purchases. In the aggregate, intangible services account for 3.9 percent of firms' total input purchases, compared with 6.4 percent in Belgium (Bijnens *et al.*, 2025).[5] A notable difference is the extensive margin. In our sample, about 50 percent of firm-years report zero intangible purchases, compared with only 18.6 percent in the Belgian data. Conditional on positive intangible purchases, the median share is 1.7 percent, with substantial right-skew (75th percentile: 6.3 percent; 90th percentile: 19.0 percent).

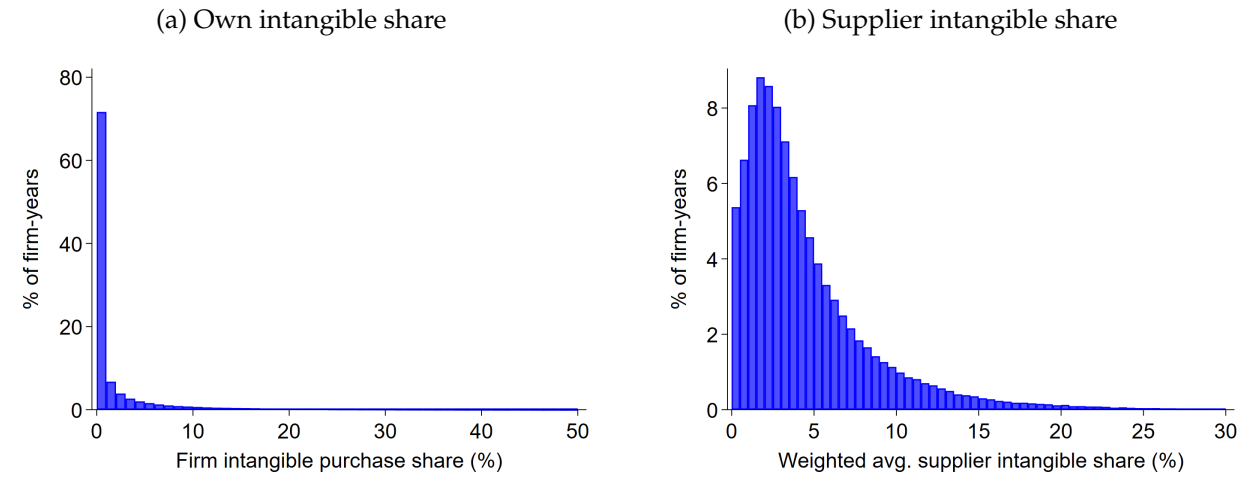
Two caveats apply to our measure. First, our data capture outsourced intangible services purchased through the VAT system but not in-house intangible production (e.g., a firm's own IT department developing software internally). To the extent that some firms with zero outsourced purchases produce intangibles in-house, we overstate the true non-adoption rate. However, this concern is mitigated by the fact that in-house intangible production requires specialized human

[5] This aggregate share equals the purchase-weighted average of the firm-level intangible shares.

capital that may be more common in advanced economies than in an emerging market like the Dominican Republic. Second, some categories in the intangible classification (e.g., personnel provision services, CIIU 7491) may capture operational expenditures rather than investment in durable intangible capital. We follow the classification of Bijnens *et al.* (2025), who include this category as part of economic competencies following the Corrado *et al.* (2005) taxonomy. Reassuringly, personnel provision accounts for less than 5 percent of total intangible purchases in our data, so excluding it would not materially affect the results.

Figure 1 illustrates the distributions of the two key intangible variables. Panel (a) shows the firm-level intangible purchase share (s_{it}^{intang} from Equation 1). The distribution is heavily right-skewed, with about 50 percent of firm-years at zero and a long tail extending beyond 50 percent. Panel (b) shows the purchase-weighted average intangible share of each firm's suppliers ($\bar{s}_{it}^{\text{suppliers}}$ from Equation 2). The averaging across suppliers produces a much smoother distribution, with the mode around 1–2 percent and substantial variation.

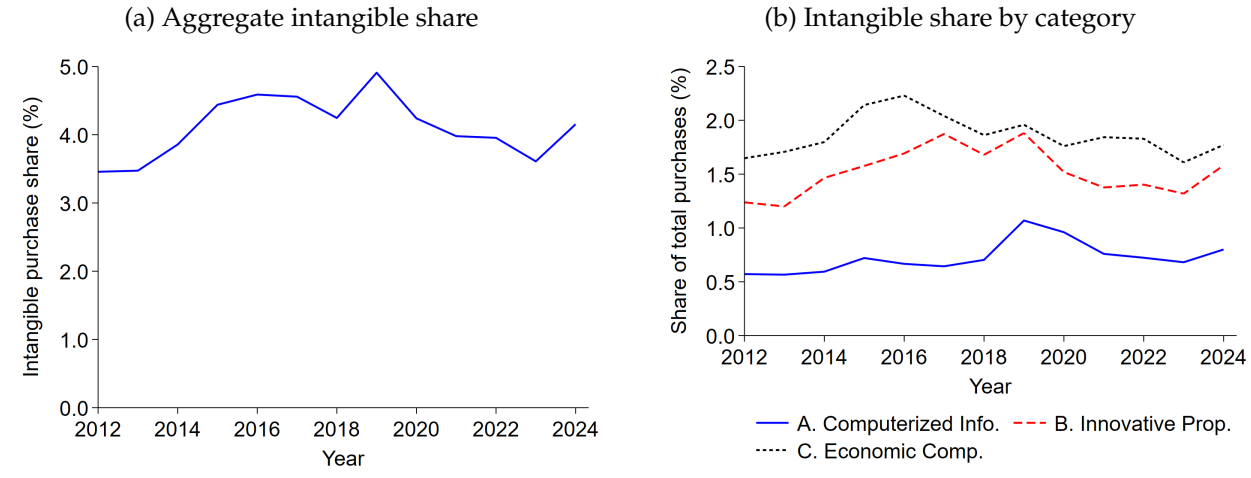
Figure 1: Distributions of intangible purchase shares



Notes: Panel (a) plots the distribution of firm-level intangible purchase share. Panel (b) plots distribution of purchase-weighted average intangible share of suppliers. The share ranges from 0 to 100%; for readability the horizontal axis is restricted to 0–50% in panel (a) and 0–30% in panel (b), and the firm-years above these cutoffs (1.8% of firm-years in panel (a) and 0.7% in panel (b)) are omitted.

Figure 2 shows the evolution of the aggregate intangible purchase share over time. Panel (a) plots the total intangible share, which rises from 3.5 percent in 2012 to a peak of approximately 4.9 percent in 2019, before declining after 2020. Panel (b) decomposes this share into the three categories of the Corrado et al. taxonomy. Economic competencies (consulting, training, advertising) accounts for the largest share, followed by innovative property (R&D, design, entertainment) and computerized information (software, data processing).

Figure 2: Intangible purchase shares over time



Notes: Panel (a) shows the purchase-weighted average intangible share across all firms with positive purchase. Panel (b) decomposes the total into: A = software + databases (blue, solid); B = R&D + entertainment + design (red, dashed); C = advertising + training/organizational capital (black, short-dashed). Categories follow the Corrado *et al.* (2005, 2009) taxonomy as implemented by Bijnens *et al.* (2025).

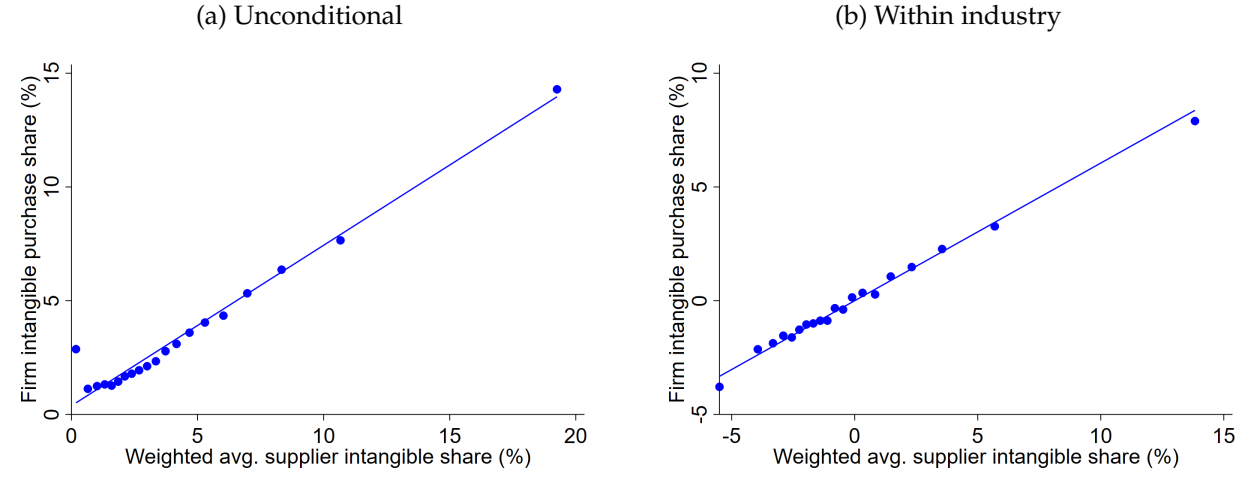
3.2 Intangible homophily in supply-chain matching

We now turn to the central question for this paper, which is whether intangible purchases exhibit complementarities along supply chains. Figure 3 provides initial evidence by plotting the relationship between a firm's own intangible purchase share and the purchase-weighted average intangible share of its suppliers, both averaged over all years. Panel (a) shows the unconditional relationship. The correlation is positive (0.33), indicating that firms purchasing more intangible inputs tend to source from suppliers that also purchase more of them. This correlation could reflect industry composition—certain industries may be inherently more intangible-intensive at all stages of production. To address this, Panel (b) plots the same relationship after residualizing both variables on 4-digit ISIC Rev. 3.1 industry fixed effects. The within-industry correlation remains positive and significant (0.23), indicating that the co-movement of intangible intensity along supply chains is not simply an artifact of sectoral sorting.

The binscatter correlations control for industry composition but not for the many firm-level characteristics—scale, productivity, demand—that could drive both a firm's intangible spending and its choice of partners. A sharper way to document sorting on intangibility is to ask whether, within the production network, intangible-using firms direct disproportionate trade to one another. Following the Khwaja and Mian (2008) two-way fixed-effects design, we test for assortative matching on intangibility at the supplier–buyer level. We collapse the firm-to-firm transaction panel to the supplier i, buyer j, year t level and estimate

$$\ln \text{value}_{ijt} = \beta \left(H_i^S \times H_j^B \right) + \alpha_{it} + \delta_{jt} + \varepsilon_{ijt}, \tag{3}$$

Figure 3: Firm intangible purchase share vs. supplier intangible purchase share



Notes: Each dot represents equal-sized bins of firms. Panel (a) shows the unconditional relationship. Panel (b) residualizes both variables on 4-digit ISIC Rev. 3.1 industry fixed effects.

where H_i^S and H_j^B are indicators that the supplier and buyer had positive intangible purchases in 2013, α_{it} are supplier-by-year fixed effects, δ_{jt} are buyer-by-year fixed effects, and standard errors are two-way clustered by supplier and buyer. The two-way fixed effects absorb every time-varying firm characteristic on each side of the link—scale, productivity, demand, prices, and common shocks—so β is identified purely from within-supplier variation across buyers of differing intangibility, and symmetrically. It therefore rules out the most natural confound—that intangible-using firms trade disproportionately with one another simply because they are also larger, more productive, or exposed to common demand—since every such firm-by-year characteristic on both sides of the link is differenced out, leaving β to capture sorting on intangibility itself.

Table 1 reports the estimates. The results in column (1) indicate that, net of every firm-by-year confounder on both sides of the link, a high-intangible supplier directs approximately 21 percent more trade value to a high-intangible buyer than a low-intangible supplier directs to that same buyer. The continuous specification in column (2) replaces the indicators with the product of the supplier's and buyer's predetermined 2013 intangible shares, $s_i^S \times s_j^B$, each scaled between zero and one. The estimate indicates that sorting operates on the intensity of intangible use beyond the adoption margin. For example, a pair in which both partners have shares of 19 percent (the 90th percentile among adopters) trades about 13 percent more than an otherwise identical pair with zero intangible shares.

The homophily premium is a cross-sectional fact about who trades with whom, net of firm-level confounders; it does not by itself identify a causal complementarity in investment. But it documents that intangible-using firms are tightly clustered together in the supply chain—the matching pattern one would expect if intangible input decisions reinforce one another across connected firms. Whether they do is the question the next two sections take up with explicit sources

of identifying variation.

Table 1: Intangible homophily in supply-chain matching

	Binary (1)	Continuous (2)
$H_i^S \times H_j^B$	0.193***	
	(0.021)	
$s_i^S \times s_j^B$		3.719***
		(0.361)
Supplier × year fixed effects	Yes	Yes
Buyer × year fixed effects	Yes	Yes
Suppliers	82,035	82,035
Pair-years	27,837,011	27,837,011

Notes: The dependent variable is the log of the annual trade value from supplier i to buyer j. Column (1) uses the interaction of binary predetermined-intangibility indicators, equal one if the firm had positive intangible purchases in 2013. Column (2) uses the interaction of the supplier's and buyer's predetermined 2013 intangible purchase shares, each scaled between zero and one. All regressions include supplier-by-year and buyer-by-year fixed effects, which absorb all time-varying firm-level characteristics on each side of the link. Standard errors two-way clustered by supplier and buyer are in parentheses.

4 Intangible input complementarities along the supply chain

We now test empirically whether complementarities in intangible purchases along the supply chain drive purchases of intangible inputs. Our identification strategy exploits variation in the composition of demand shocks received by supplier firms. The idea is that each supplier receives demand shocks from its buyers through the production network. We decompose these shocks according to the intangible intensity of the buyer, and test whether shocks from buyers that themselves purchase intangibles elicit a stronger intangible purchase response from the supplier than shocks from buyers that do not.

Following Dhyne *et al.* (2021), we construct a firm-level export-demand shock driven entirely by changes in world demand for the precise products that Dominican firms export. For each exporting firm f in year t,

$$z_{ft}^{\text{direct}} = \left(\frac{\text{exports}_{f,t-1}}{\text{revenue}_{f,t-1}} \right) \sum_{c,h} \omega_{f,ch,t-1} \, \Delta \ln \text{WID}_{c,h,t}^{-DR}, \tag{4}$$

where $\omega_{f,ch,t-1}$ is the share of firm f's exports going to destination country c in product h in year $t-1$, and $\text{WID}_{c,h,t}^{-DR}$ is world import demand for product h by country c excluding the Dominican Republic, computed from the BACI reconciled bilateral trade data (Gaulier and Zignago, 2010). The leading factor scales the shock by the firm's lagged export intensity, with exports converted to

pesos at the annual exchange rate. Products are defined at the six-digit Harmonized System level, which matches the foreign-demand shifter to the precise good the firm exports.[6]

We propagate these direct shocks through the domestic production network using a Leontief-type inverse to obtain each firm's total (direct plus network-transmitted) demand shock:

$$\mathbf{z}_t^{\text{total}} = (\mathbf{I} - \mathbf{\Omega}_{t-1})^{-1} \mathbf{z}_t^{\text{direct}}, \tag{5}$$

where $\Omega_{ij,t-1} = \text{sales}_{ij,t-1}/\text{revenue}_{i,t-1}$ is the firm-level sales-share matrix and $\mathbf{I}$ is the identity. The lagged structure ensures that the network through which shocks propagate is predetermined relative to the shocks themselves.[7]

For each supplier i in year t, we decompose its buyer-side demand exposure into two components by the intangible status of each buyer. Let $\mathcal{H}$ denote the set of buyers with positive intangible purchases in 2013—a predetermined classification, since the foreign-demand shock is first defined in 2014, so the split cannot respond to any shock we exploit—and let $w_{ijt} = \text{sales}_{ij,t} / \sum_k \text{sales}_{ik,t}$ be the share of supplier i's sales going to buyer j. We construct

$$z_{it}^H = \sum_{j \in \mathcal{H}} w_{ijt} \cdot z_{jt}^{\text{total}}, \tag{6}$$

$$z_{it}^L = \sum_{j \notin \mathcal{H}} w_{ijt} \cdot z_{jt}^{\text{total}}. \tag{7}$$

z_{it}^H is the supplier's sales-weighted exposure to foreign demand transmitted through intangible-purchasing buyers; z_{it}^L is the analogous exposure through non-intangible buyers. By construction, $z_{it}^H + z_{it}^L = \sum_j w_{ijt} z_{jt}^{\text{total}}$, so the two components exhaust the supplier's total buyer-side shock exposure.

Intangible purchases are generally lumpy and may change slowly, so we estimate its dynamic response to these exposures with local projections. For each horizon $\kappa \in \{-3, \ldots, 6\}$ we regress the cumulative change in supplier i's intangible purchase share between the pre-shock year $t-1$ and year $t+\kappa$ on the two exposure components,

$$s_{i,t+\kappa} - s_{i,t-1} = \beta_H^\kappa z_{it}^H + \beta_L^\kappa z_{it}^L + \mathbf{X}'_{i,t-1}\delta^\kappa + \alpha_i^\kappa + \mu_{jt}^\kappa + \varepsilon_{it}^\kappa, \tag{8}$$

where s_{it} is the intangible share of total purchases, $\mathbf{X}_{i,t-1}$ is a vector of one-period-lagged controls (log number of suppliers, log sales, and log employment), α_i^κ are firm fixed effects, and μ_{jt}^κ are industry $\times$ year fixed effects. The coefficient β_H^κ and β_L^κ trace the cumulative response of the intangible share to a demand shock transmitted through high- and low-intangible buyers, respectively. Because the dependent variable is the cumulative change in the share relative to $t-1$,

[6] Two features of the Dominican setting make this shock demanding of the data. First, the export base is narrow—about 2,000 firms have non-negligible exports in a typical year—so the direct shock is defined for a small set of firms before propagation. Second, the intangible-purchasing firms whose complementarity we study are concentrated in non-tradable services, reached only after the shock propagates upstream through the domestic network.

[7] We present a set of validation tests of $\mathbf{z}_t^{\text{total}}$ as a firm-level demand shock in Appendix A-I.

the coefficient at $\kappa = -1$ is identically zero by construction; each estimate therefore measures the cumulative change in the intangible share relative to the year before the shock, and the pre-shock horizons ($\kappa < -1$) provide a test of parallel trends. Under the complementarity hypothesis, $\beta_H^\kappa > 0$ and grows with the horizon as the lumpy investment is realized. Standard errors are clustered at the firm level.

The estimates in Figure 4 point to supply-chain complementarities in intangible purchases. Panel (a) plots the cumulative response of the intangible purchase share to demand transmitted through intangible-purchasing buyers. The pre-shock coefficients are small and statistically indistinguishable from zero, consistent with parallel trends, and the on-impact response is also imprecise. The effect then builds steadily with the horizon and becomes statistically significant three years after the shock. The intangible purchase share rises by 1.3 percentage points three years after the shock, by 1.7 percentage points after four years, and by 3.1 percentage points after five years.[8] Panel (b) plots the analogous response to demand transmitted through other buyers. The response is flat, centered on zero, and statistically insignificant at virtually every horizon. Based on the estimates in Table A1, a one-unit shock raises the recipient firm's sales by about 15 percent on impact. For a supplier selling its entire output to intangible-purchasing buyers hit by such a shock, the intangible purchase share rises by 3.1 percentage points over five years, whereas demand routed through other buyers has no effect at any horizon.[9] We conclude that demand shocks routed through intangible-intensive buyers produce a gradual, persistent deepening of the supplier's own intangible intensity, whereas demand through other buyers does not.

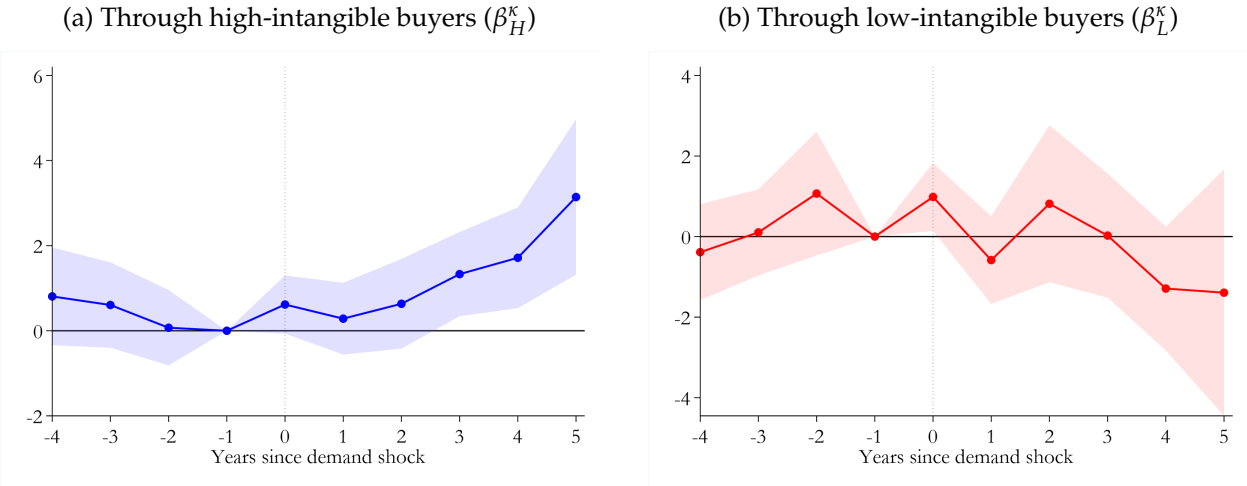
Appendix A-II documents a cross-country analogue of this pattern, mapping exporter countries to suppliers and destination countries to buyers. Countries whose exports were directed toward intangible-intensive destinations subsequently accumulated more intangible capital. Among advanced economies, a ten-percentage-point higher predetermined exposure is associated with a 1.3-percentage-point larger increase in the intangible-investment share of GDP over the following decade, and the pattern survives in a broader sample of 201 countries built from bilateral trade data alone, where the outcome is the intangible content of a country's imports. The exercise is cross-sectional and descriptive, but it indicates that the mechanism we identify at the firm level leaves the same footprint in aggregate data.

Robustness Table 2 examines the robustness of these results along several dimensions. Each column re-estimates the local projection Equation 8, varying one feature of the baseline, and reports the high- and low-intangible-buyer coefficients at the five-year horizon, which is where the response is largest. The baseline estimate (column 1) is larger when the foreign-demand shock is winsorized at the 1st and 99th percentiles before propagation (column 2), so the result is not

[8] The persistent response is not driven by the shocks' serial correlation. The foreign-demand shock has a first-order autocorrelation of 0.01 in the pooled panel and -0.14 within firm.

[9] The standard deviations of z_{it}^H and z_{it}^L are 0.035 and 0.022, respectively; the distributions are heavily right-skewed, with most suppliers having small exposures and a small set of export-connected suppliers carrying large shocks. A one-standard-deviation increase in z_{it}^H therefore raises the intangible purchase share by about 0.11 percentage points after five years, roughly 3 percent of the average intangible purchase share.

Figure 4: Response of intangible purchase share to demand shocks



Notes: The figure shows the response of the intangible purchase share to a foreign demand shock transmitted through high-intangible buyers (panel a) and low-intangible buyers (panel b). Buyers are classified as high-intangible if they had positive intangible purchases in 2013 (predetermined, since the shock is first defined in 2014). All regressions include firm fixed effects, six-digit CIIU industry × year fixed effects, and one-period lagged controls (log number of suppliers, log sales, and log employment). Shaded areas denote the 90 percent confidence intervals, constructed from standard errors clustered at the firm level.

driven by extreme shock realizations. Varying the minimum ratio of business-to-business sales to revenue that defines the estimation sample—from the baseline 0.10 to a looser 0.05 (column 3) or a stricter 0.25 (column 4)—leaves the coefficient stable, indicating that it is not sensitive to the treatment of firms with thin firm-to-firm linkages. Restricting to a balanced panel of suppliers observed at every horizon $\kappa = 0, \ldots, 5$ yields a slightly larger coefficient (column 5), so the build-up in Figure 4 is not an artifact of differential attrition across horizons. Finally, a placebo that randomly permutes the supplier exposures within each year returns a statistically insignificant coefficient (column 6). Across all columns the low-intangible-buyer coefficient is statistically indistinguishable from zero.

4.1 COVID-19 as a natural demand shock to tourism

We now leverage the COVID-19 collapse of Dominican tourism as a quasi-experimental source of demand variation. The pandemic delivered a large, common, and unambiguously negative demand shock to firms in the accommodation and food-service sector, and the shock propagated to those firms' suppliers through the production network. Under the cost-reducing complementarity mechanism, the shock should pass through more strongly to suppliers whose tourism buyers are themselves intangible-using than to suppliers whose tourism buyers are not.

The cost-reducing complementarity mechanism predicts that a transitory demand shock to a supplier's buyers will reduce the supplier's intangible purchases for the duration of the shock and revert to baseline once buyers recover. The mechanism, however, operates on different margins

Table 2: Robustness of the intangible-share response

	Baseline	Shock winsor. 1/99	B2B ≥ 0.05	B2B ≥ 0.25	Balanced panel	Placebo
	(1)	(2)	(3)	(4)	(5)	(6)
z_{it}^{H}	3.143***	4.017***	3.162***	3.071***	3.490***	−1.388
	(1.110)	(1.349)	(1.216)	(1.191)	(1.129)	(1.671)
z_{it}^{L}	−1.393	−0.810	−1.354	−2.170	−1.360	0.549
	(1.859)	(2.282)	(1.767)	(2.060)	(2.004)	(1.768)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Within R^2	0.001	0.001	0.001	0.001	0.001	0.001
Observations	80,808	80,808	87,403	68,388	76,914	80,808

Notes: Each column reports the coefficients on the high- and low-intangible-buyer exposures estimated at the five-year horizon. The dependent variable is the cumulative change in the intangible purchase share between $t-1$ and $t+5$. All specifications include firm and six-digit CIIU industry × year fixed effects and the lagged controls of the baseline. Column (1) is the baseline (raw foreign-demand shock, B2B-to-revenue ratio ≥ 0.10). Column (2) winsorizes the buyer-level foreign-demand shock at the 1st and 99th percentiles by year before network propagation. Columns (3) and (4) change the minimum B2B-to-revenue ratio to 0.05 and 0.25. Column (5) restricts to suppliers with a non-missing outcome at every horizon $\kappa = 0, \ldots, 5$. Column (6) randomly permutes the supplier exposures across firms within each year. Standard errors clustered at the firm level are in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

at different speeds. Intangible adoption and de-adoption (the extensive margin) involves fixed costs—installing software, training staff, integrating systems—that take more than a year to amortize or reverse. A supplier facing a one-year transitory demand drop has no incentive to terminate intangible relationships that took years to build, since the option value of staying connected exceeds the short-run flow cost. By contrast, intensive-margin adjustments—scaling subscription tiers, reducing consultant hours, deferring upgrades—can be made at higher frequencies without committing the firm to a permanent exit. We therefore expect a transitory shock like COVID to manifest on the intensive margin among incumbent adopters.

We define a buyer b as a 'tourism buyer' if its 1-digit ISIC industry classification is "Hotels and restaurants". This category includes hotels, motels, restaurants, bars, and catering services.[10] The intangible status of each buyer is fixed at a pre-COVID baseline, so a buyer is classified as high-intangible (H) if it had positive intangible purchases in each of the three pre-COVID years 2017, 2018, and 2019; otherwise it is classified as low-intangible (L):

$$\text{buyer}_H = \mathbf{1}\{I_{b,2017} > 0 \text{ and } I_{b,2018} > 0 \text{ and } I_{b,2019} > 0\}, \qquad \text{buyer}_L = 1 - \text{buyer}_H. \tag{9}$$

This classification is fully predetermined with respect to the pandemic and never updated thereafter. It selects, on the H side, firms that were structurally intangible-using before COVID (about 7.8 percent of all buyers, predominantly the largest hotel and restaurant chains). The L side in-

[10] We do not include air transport or travel agencies in the tourism flag because the Dominican industry codes group these with broader transport and business-service categories that contain non-tourism activity.

cludes both true non-adopters and occasional adopters whose intangible purchases were sporadic in the pre-COVID window.

For each supplier f, we compute baseline exposure shares from the VAT firm-to-firm transactions:

$$\begin{aligned} \text{TourShare}_f^H &= \sum_b \omega_{fb}^{\text{base}} \cdot \mathbf{1}\{b \in \text{tourism}\} \cdot \text{buyer}_H, \\ \text{TourShare}_f^L &= \sum_b \omega_{fb}^{\text{base}} \cdot \mathbf{1}\{b \in \text{tourism}\} \cdot \text{buyer}_L, \end{aligned} \tag{10}$$

where $\omega_{fb}^{\text{base}}$ is the supplier's sales share to buyer b averaged over 2017–2019.[11]

We then estimate the following triple-difference-in-differences specification on the sample spanning the period 2017–2024:

$$Y_{f,t} = \alpha_f + \mu_{j,t} + \delta_H \text{TourShare}_f^H \times \mathbf{1}\{t = 2020\} + \delta_L \text{TourShare}_f^L \times \mathbf{1}\{t = 2020\} + \gamma \mathbf{X}'_{f,t-1} + \varepsilon_{f,t}, \tag{11}$$

where $Y_{f,t}$ is the intensive margin of intangible use—the change in log intangible purchases $\Delta \ln I_{f,t}$ among firm-years with positive intangibles; firm fixed effects α_f absorb the time-invariant component of each supplier's intangible behavior; 4-digit industry $\times$ year fixed effects $\mu_{j,t}$ absorb common industry-time shocks (so the aggregate component of COVID's effect on the supplier's industry is differenced out, and δ_H, δ_L are identified off cross-sectional variation in tourism-buyer exposure within industry-year cells); and $\mathbf{X}_{f,t-1}$ denotes lagged controls (log number of suppliers, log sales, and log employment). Standard errors are clustered at the firm level. We focus on the impact year 2020 because this is the year of the largest tourism collapse.

Under the cost-reducing complementarity hypothesis, COVID's negative shock to tourism should transmit through the H channel (intangible-using tourism buyers) but not through the L channel. Table 3 reports the results for the change in log intangible purchases among incumbent adopters for the pandemic year (2020), the year after, and a pre-pandemic placebo year (2018). In the pandemic year (column 1), suppliers more exposed to high-intangible tourism buyers sharply cut their intangible purchases. Specifically, a supplier that sold its entire output to high-intangible tourism buyers reduced its intangible purchases by about 84 percent in 2020, relative to otherwise-similar firms, while exposure to low-intangible tourism buyers had a small, statistically insignificant effect. The timing of the effect is informative. By 2021 (column 2), as tourism began to recover, $\hat{\delta}_H$ becomes statistically indistinguishable from zero—consistent with the model's prediction that a transitory demand shock produces a transitory response and leaves no persistent reduction in supplier intangible purchases. A pre-pandemic placebo that re-times the entire design to 2018—

[11] Across 68,254 suppliers with non-missing baseline (2017–2019) sales, mean H-tourism exposure is 1.7 percent and mean L-tourism exposure is 0.6 percent. The cross-supplier distribution of these tourism-exposure shares is heavily right-skewed—896 suppliers sell more than half of their output to tourism buyers, and the top percentile sell essentially all of it. The strict-H classification leaves 92 percent of buyers in the L group; alternative classifications produce qualitatively similar results but quantitatively different estimates of δ_L.

classifying buyers on 2015–2017 and computing tourism exposure from a 2015–2017 baseline over a 2015–2019 window—likewise yields an insignificant coefficient.[12]

Table 3: COVID-19 as a tourism demand shock

	(1) Post = 2020	(2) Post = 2021	(3) Placebo (2018)
$\hat{\delta}_H$	-1.829^{***}	0.300	0.181
	(0.533)	(0.612)	(0.774)
$\hat{\delta}_L$	0.507	0.054	0.247
	(1.416)	(2.604)	(1.385)
Wald test $\delta_H = \delta_L$ $[F, p]$	2.31 [0.128]	0.01 [0.927]	0.00 [0.967]
Within R^2	0.015	0.015	0.018
N	68,123	68,123	32,367

Notes: The dependent variable is the change in log intangible purchases for firm-years with positive intangibles. δ_H and δ_L are the coefficients of supplier's baseline sales shares to H- and L-tourism buyers, where a tourism buyer is H if it had positive intangible purchases in each of the three pre-shock baseline years and L otherwise. Columns (1) and (2) classify buyers on 2017–2019 and use the 2017–2024 sample, with the post indicator taking value one in 2020 and 2021, respectively. Column (3) is a pre-pandemic placebo, for which buyers are classified on 2015–2017, tourism exposure is computed from 2015–2017 transactions, the sample is 2015–2019, and the post indicator is equal one in 2018. All regressions include firm fixed effects, 4-digit industry × year fixed effects, and lagged controls (log number of suppliers, log sales, and log employment). Standard errors clustered at the firm level are in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

These estimates help separate the complementarity from a pure information channel. A natural alternative reading of buyer-supplier correlations in intangible purchases is learning, which implies that suppliers discover intangible inputs from the firms they trade with. The empirical exercises presented in this section mitigate this concern. We argue that, because every supplier in the estimation sample is already connected to its buyers, whatever information flows across these relationships flows with or without the shock, and the estimates therefore isolate the complementarity. While it could still be argued that the flow of information is stronger the higher the intensity of existing relationships, the tourism collapse speaks against this. This is because suppliers exposed to intangible-using tourism buyers cut their intangible purchases sharply in 2020 and reverted as tourism recovered. Since information, once transmitted, is not withdrawn when a buyer's demand collapses, the reversal is consistent with a complementarity that prices current demand.

[12] Because TourShare^L is tiny and imprecisely estimated for most suppliers, the formal $\delta_H = \delta_L$ test for 2020 is borderline insignificant, with a p-value of 0.13.; yet, the evidence is carried by the large, precisely estimated $\hat{\delta}_H$ and, above all, by its timing.

5 Linkages to intangible-using buyers and intangible-producing workers

The previous section identified the complementarity by holding relationships fixed, so that information flowing between partners was netted out. Information, however, is part of how the network operates. The formation of supply relationships (Alfaro-Ureña *et al.*, 2021; Amiti *et al.*, 2024; Criscuolo *et al.*, 2025) and worker flows between firms (Stoyanov and Zubanov, 2012; Jarosch *et al.*, 2021; Herkenhoff *et al.*, 2024) are established channels of knowledge diffusion. This section presents two event studies on the extensive margin of intangible use. We first ask whether a supplier that starts selling to an intangible-using buyer becomes more likely to purchase intangible inputs itself—the total effect of a new relationship, which bundles the demand incentive of section 4 with the information the new partner brings. Second, building on the evidence of supply chain-specific human capital (Cardoza *et al.*, 2025), we ask whether hiring from an intangible producer has a similar effect in the sense that such workers carry the know-how needed to put intangible inputs to productive use.

Both exercises adopt a stacked event-study design (Cengiz *et al.*, 2019; Goodman-Bacon, 2021; Sun and Abraham, 2021). Let t_i^* denote firm i's treatment year—the year of its first exposure to an intangible-producing partner—and define a cohort c as the set of firms that share the same treatment year. For each cohort, we construct a separate sub-dataset containing its treated firms ($t_i^* = c$) together with control firms that experience an event of the same type at c—a new buyer link or a new hire—but with a partner outside the intangible-producing sector, and whose own first intangible-producing exposure (if any) occurs after $c + 4$; each sub-dataset covers the event window $[c - 4, c + 4]$. Because treated and control firms experience the same type of event at the same time, the comparison isolates the intangibility of the new partner rather than the event itself, while firms retain any prior intangible-purchase history.[13] Stacking the cohort-specific sub-datasets, we estimate:

$$Y_{it} = \sum_{k=-4}^{4} \beta_k \cdot \mathbf{1}\{t - t_i^* = k\} \times D_i + \gamma' X_{i,t_i^*-1} + \alpha_{i,c} + \delta_{c,t} + \eta_{s(i),t} + \varepsilon_{it}, \tag{12}$$

where Y_{it} is an indicator taking value one if the firms makes any intangible purchase—the extensive margin of intangible adoption.[14] D_i is an indicator for treated firms, X_{i,t_i^*-1} includes log number of suppliers, log sales, and log employment measured at the pre-treatment baseline period $t_i^* - 1$ (held fixed across all event-time periods to avoid conditioning on post-treatment outcomes), $\alpha_{i,c}$ are cohort-specific firm fixed effects, $\delta_{c,t}$ are cohort $\times$ year fixed effects, $\eta_{s(i),t}$ are 4-digit indus-

[13] Pre-event coefficients are therefore not mechanically zero and constitute a genuine test of parallel trends.

[14] We focus on the extensive margin rather than the intangible purchase share because the denominator of the latter—total purchases—itself responds to hiring and linking events, so a change in the share would confound intangible adoption with movements in the firm's overall purchasing, whereas the binary adoption indicator is invariant to the denominator. The share remains the appropriate outcome for the demand-shock local projection in Figure 4, which is estimated on the full panel with continuous exposure and is designed to trace the gradual deepening of intangible intensity rather than a discrete transition into adoption.

try ($s(i)$) $\times$ year fixed effects, and $k = -1$ is the reference period. Standard errors are clustered at the firm level.

We begin with the formation of new buyer-supplier relationships, asking whether a supply-chain link with an intangible-intensive buyer triggers the supplier to begin purchasing intangible inputs. Using the VAT transaction data, we identify the first year in which a supplier forms a transaction link with a buyer classified as an intangible-producing industry.[15] Treated firms are those whose first such link occurs at t; controls form a new link with a non-intangible-industry buyer at t and have no intangible-industry link until after $t + 4$. The matched new-relationship requirement is important here because linking to an intangible-industry buyer is correlated with a firm's broader adoption trajectory; conditioning both groups on forming a new relationship at t removes that trajectory from the comparison.

Panel (a) of Figure 5 reports the results. The pre-event coefficients are statistically insignificant except the one at $k = -4$, which retains a small, marginally significant negative value. At $k = 0$—the year the supplier first sells to an intangible-producing buyer—the probability of any intangible purchase rises by 4.0 percentage points, and the effect grows to roughly 5–6 percentage points through $k = 4$, all statistically significant. These results show that acquiring an intangible-intensive buyer moves the supplier across the adoption threshold. That is, when a firm begins selling to a buyer in an intangible-producing industry, it starts purchasing intangible services itself.

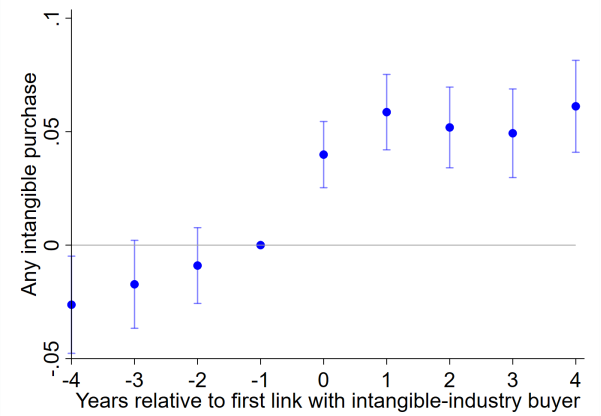
We then turn to hiring, which tests whether knowledge transfer through worker mobility can push firms across the same threshold. Supply chains transmit knowledge as well as demand, notably through the workers who move along them (Cardoza *et al.*, 2025). We then ask whether such movements matter for adoption. A newly hired worker brings know-how but not necessarily demand, isolating the capability side of the threshold. Leveraging the employer-employee matched data, we track all worker moves between firms in consecutive years and classify each move by whether the worker's previous employer was in an intangible-producing industry. The sample consists of firms that had never hired from an intangible-producing firm before year t; treated firms are those whose first such hire occurs at t, and controls hire from a non-intangible-producing firm at t, with their first intangible-producing hire (if any) after $t + 4$. We estimate Equation 12 with the same stacked design and the extensive-margin outcome.

Panel (b) of Figure 5 plots the event-study coefficients. The pre-event coefficients are small and statistically indistinguishable from zero, consistent with parallel trends. At $k = 0$—the year of the first hire from an intangible-producing firm—the probability of any intangible purchase rises by 4.1 percentage points. The effect is statistically significant and persistent, stabilizing around 3–4 percentage points 4 years after the hire. The magnitude on impact is comparable to the new-link effect, indicating that both forms of first exposure operate primarily by pushing firms across the adoption threshold. The result also resonates with the learning-from-coworkers channel in

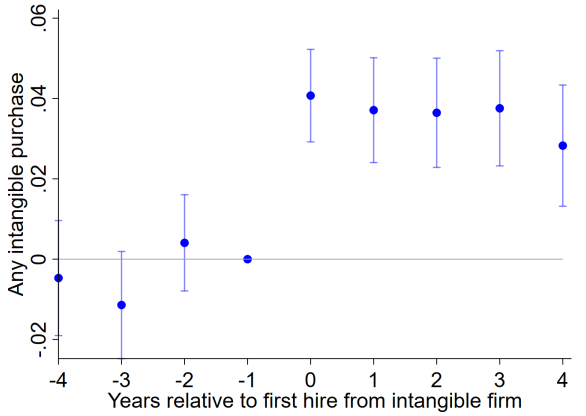
[15] We use the industry-based classification because it is predetermined and not subject to the same endogeneity concerns as purchase-based measures.

Figure 5: Intangible adoption around new exposure to intangible-intensive partners

(a) First link with an intangible-industry buyer



(b) First hire from an intangible-producing firm



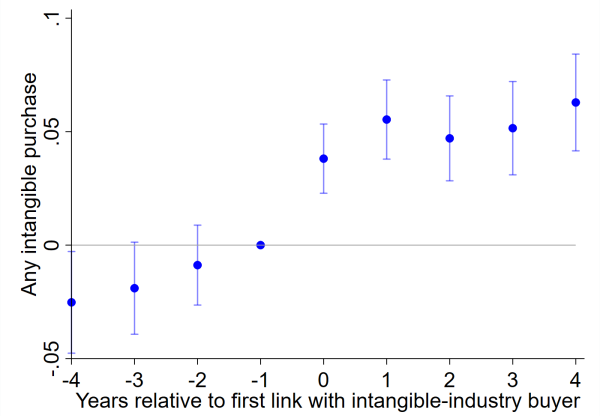
Notes: The dependent variable is an indicator for any intangible purchase. Panel (a) shows the event study coefficients around the first transaction link with a buyer in an intangible-producing industry; treated firms are first-linkers at t, controls form a new link with a non-intangible-industry buyer at t and have no intangible-industry link until after $t+4$. Panel (b) shows the event study coefficients around the first hire of a worker from an intangible-producing firm; treated firms are first-hirers at t, controls are firms that hire from a non-intangible-producing firm at t and whose first intangible-producing hire (if any) occurs after $t+4$. All regressions include cohort-specific firm fixed effects, cohort $\times$ year fixed effects, and 4-digit industry $\times$ year fixed effects, and controls as of the year prior to the event (log number of suppliers, log sales, and log employment). The vertical lines denote the 90% confidence intervals, constructed from standard errors clustered at the firm level.

Jarosch *et al.* (2021)—the newly hired worker transmits tacit knowledge about intangible-intensive production processes to the firm's existing workforce, creating a durable shift in the firm's technological capacity.

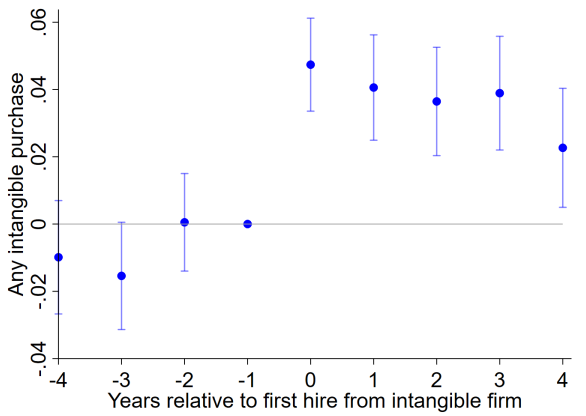
Robustness A natural concern for the event-study results is that firms in intangible-producing industries (software, R&D, consulting, and related sectors) mechanically purchase intangibles as intermediate inputs rather than as genuine investments. We therefore replicate both event studies—with the same extensive-margin specifications, including the matched-event control for the new-link design—excluding these suppliers. Figure 6 reports the results, which suggest that the effects and pre-trend patterns are similar to the full-sample figure, confirming that the event-study evidence is not driven by intangible-producing firms.

Figure 6: Event studies excluding intangible-producing suppliers

(a) First link with an intangible-industry buyer



(b) First hire from an intangible-producing firm



Notes: The dependent variable is an indicator for any intangible purchase. Panel (a) shows the event study coefficients around the first transaction link with a buyer in an intangible-producing industry; treated firms are first-linkers at t, controls form a new link with a non-intangible-industry buyer at t and have no intangible-industry link until after $t+4$. Panel (b) shows the event study coefficients around the first hire of a worker from an intangible-producing firm; treated firms are first-hirers at t, controls are firms that hire from a non-intangible-producing firm at t and whose first intangible-producing hire (if any) occurs after $t+4$. The sample excludes suppliers in intangible-producing industries. All regressions include cohort-specific firm fixed effects, cohort $\times$ year fixed effects, and 4-digit industry $\times$ year fixed effects, and controls as of the year prior to the event (log number of suppliers, log sales, and log employment). The vertical lines denote the 90% confidence intervals, constructed from standard errors clustered at the firm level.

6 A model of intangible complementarities in production networks

The empirical results establish three channels through which the use of intangible inputs propagates along supply chains. These are demand-driven complementarity, knowledge transfer through hiring, and buyer-link formation. To quantify the aggregate importance of these channels, we develop a general equilibrium model of a production network with three key ingredients: (i) endogenous intangible input use with supply-chain complementarities, (ii) an intangible-producing sector with endogenous prices, and (iii) a frictional labor market in which workers carry knowledge between firms.

6.1 Environment

The economy consists of N firms indexed by i, connected through a production network $\mathbf{\Omega} = \{\omega_{ij}\}$ observed in the VAT data, a representative household, and a continuum of workers of measure $\bar{L}$. A subset $\mathcal{I} \subset \{1, \ldots, N\}$ of firms are classified as intangible producers (software, R&D, consulting) based on their CIIU industry codes.

Household The representative household has discount factor β and consumes a CES aggregate of firm outputs:

$$C = \left(\sum_{i=1}^{N} \gamma_i^{1/\varepsilon} \, c_i^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}}, \tag{13}$$

where γ_i captures expenditure weights and ε is the elasticity of substitution across varieties. The household supplies $\bar{L}$ units of labor inelastically, owns all firms, and maximizes utility subject to the budget constraint $\sum_i p_i \, c_i \leq \sum_i w_i L_i + \Pi$, where w_i is the bargained wage at firm i and Π denotes aggregate profits. In steady state, the real interest rate satisfies $r = 1/\beta - 1$.

Production technology Each firm i produces using labor L_i, an intermediate composite M_i, and a flow of intangible services U_i:

$$Y_i = A_i \left[\alpha_{Li} \, L_i^{\frac{\sigma-1}{\sigma}} + \alpha_{Mi} \, M_i^{\frac{\sigma-1}{\sigma}} + \alpha_{Ui} \, U_i^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{14}$$

where A_i is firm-level productivity (normalized to 1 in the quantitative model), $\sigma < 1$ governs the elasticity of substitution among inputs (so that inputs are complements), and $\alpha_{Li} + \alpha_{Mi} + \alpha_{Ui} = 1$. The intermediate composite aggregates purchases from other firms:

$$M_i = \left(\sum_{j=1}^{N} \omega_{ij}^{1/\eta} \, x_{ij}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \tag{15}$$

where ω_{ij} are observed from the VAT transaction data.

6.2 Intangible inputs and supply-chain complementarities

The key mechanism is that the effective cost of intangible services for a supplier depends on the intangible intensity of its buyers. Define the buyer-weighted intangible intensity of firm i:

$$\bar{U}_i^b = \sum_j s_{ij} \, \frac{U_j}{Y_j}, \tag{16}$$

where s_{ij} is the share of firm i's sales going to buyer j and U_j/Y_j is buyer j's intangible intensity. Intangible services are a flow input, purchased each period from the intangible-producing sector $\mathcal{I}$ at price p_I; this matches the measurement in the data, which records purchases of intangible services rather than a capital stock. The complementarity operates through the effective price of these services. That is, for a firm whose buyers are intangible-intensive, implementing intangible services is cheaper because of shared digital standards, compatible data systems, and knowledge

spillovers from buyers. Formally, the effective price of intangible services for firm i is:

$$c_{Ui} = \frac{p_I}{1 + \theta\, \bar{U}_i^b}, \tag{17}$$

where $\theta \geq 0$ is the complementarity parameter. When $\theta > 0$, a higher buyer intangible intensity $\bar{U}_i^b$ reduces the cost of using intangible services, increasing both the intensive and extensive margins of intangible use.

6.3 Intangible-producing sector

Firms in $\mathcal{I}$ produce intangible services using a technology that exhibits decreasing returns to variable inputs:

$$q_j = B_j\, K_j^{1-\nu}\, \ell_j^{\nu}, \qquad j \in \mathcal{I}, \tag{18}$$

where K_j is a sector-specific fixed factor (proprietary knowledge, specialized infrastructure), ℓ_j is labor allocated to intangible production, $\nu < 1$ governs the degree of decreasing returns, and B_j is productivity. The fixed factor creates an upward-sloping supply curve for intangible services, such that as aggregate demand for intangibles rises, the marginal cost of production increases, bidding up p_I. The aggregate supply of intangible services is:

$$Q_I = \left(\sum_{j \in \mathcal{I}} \zeta_j^{1/\rho}\, q_j^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}, \tag{19}$$

where ζ_j captures firm j's capacity weight, ρ governs the elasticity of substitution among intangible varieties, and the CES price index p_I is determined by market clearing. The decreasing returns to the variable input at the firm level ($\nu < 1$), combined with the finite number of intangible producers, generates the aggregate supply elasticity that determines how much p_I rises in response to an adoption wave.

6.4 Frictional labor market with knowledge transfer

Workers search for jobs through a Diamond-Mortensen-Pissarides matching process. The aggregate matching function is $m(u, v) = \mu\, u^{\alpha_m} v^{1-\alpha_m}$, where u is the measure of unemployed workers, v is aggregate vacancies, and μ is matching efficiency. Labor market tightness is $\xi \equiv v/u$. The job-finding rate for workers is $\lambda(\xi) = m(u, v)/u = \mu\, \xi^{1-\alpha_m}$ and the vacancy-filling rate for firms is $q(\xi) = m(u, v)/v = \mu\, \xi^{-\alpha_m}$.

Workers carry a binary knowledge type $h \in \{0, 1\}$. A worker who separates from an intangible-producing firm ($j \in \mathcal{I}$) acquires $h = 1$, representing tacit knowledge of intangible-intensive production processes. This knowledge reduces the effective cost of intangible adoption at the new

employer:

$$f_i = f \cdot (1 - \kappa\, h_i), \tag{20}$$

where $h_i \in [0, 1]$ is the share of firm i's workforce with intangible knowledge, $f > 0$ is the baseline fixed cost of intangible adoption, and $\kappa \in (0, 1)$ is the knowledge transfer parameter. A higher κ means that workers from intangible-producing firms bring more useful knowledge, reducing the reorganization and training costs that firms must incur to adopt intangibles.

Job creation and wages Each firm i posts vacancies at cost c_v per vacancy. Free entry drives the expected cost of filling a vacancy to equal the value of a filled job:

$$\frac{c_v}{q(\xi)} = J_i, \tag{21}$$

where J_i is the present value of a filled position at firm i, satisfying the asset equation

$$r\, J_i = p_i \cdot \mathrm{MPL}_i - w_i - \delta_L\, J_i, \tag{22}$$

where $\mathrm{MPL}_i = \partial Y_i / \partial L_i$ is the marginal product of labor, w_i is the wage, and δ_L is the exogenous separation rate. Wages are determined by Nash bargaining between the firm and the worker, with the worker's bargaining weight ϕ:

$$w_i = (1 - \phi)\, b + \phi\, (p_i \cdot \mathrm{MPL}_i + c_v\, \xi)\,, \tag{23}$$

where b is the flow value of unemployment. Under the Hosios condition ($\phi = \alpha_m$), the decentralized equilibrium is constrained efficient.

Steady-state knowledge distribution In steady state, employment relationships form and dissolve at rate δ_L. Workers who separate from a firm in $\mathcal{I}$ enter the unemployment pool with $h = 1$; all others enter with $h = 0$. The steady-state fraction of the unemployment pool with intangible knowledge is

$$\pi_h = \frac{L_{\mathcal{I}}}{\bar{L}}, \tag{24}$$

where $L_{\mathcal{I}} = \sum_{j \in \mathcal{I}} L_j$ is total employment in intangible-producing firms. Under random matching, each firm draws workers from this pool and $h_i = \pi_h$ for all i. In the quantitative model, we take firm-specific h_i directly from the employer-employee data, which captures deviations from random matching due to local labor markets, industry proximity, and geographic concentration of intangible producers.

6.5 Firm's problem and equilibrium

Each firm i posts vacancies v_i at flow cost c_v per vacancy and chooses intermediate inputs $\{x_{ij}\}_{j=1}^{N}$ and intangible services U_i to maximize steady-state profits:

$$\max_{\{x_{ij}\},U_i,v_i} \; p_i Y_i - w_i L_i - \sum_{j=1}^{N} p_j x_{ij} - c_{Ui} U_i - c_v v_i - f_i \cdot \mathbf{1}\{U_i > 0\}, \tag{25}$$

where w_i is the bargained wage from Equation 23, $c_{Ui} = p_I/(1+\theta \bar{U}_i^b)$ is the effective price of intangible services from Equation 17, and $f_i = f(1-\kappa h_i)$ is the knowledge-adjusted fixed cost from Equation 20. Employment L_i is not a static choice but is determined by the steady-state flow balance $\delta_L L_i = q(\xi) v_i$, where $q(\xi)$ is the vacancy-filling rate. The firm's intangible input decision has two margins.

On the intensive margin, the first-order condition for U_i equates the marginal revenue product to the effective price:

$$\frac{p_I}{1+\theta\,\bar{U}_i^b} = p_i\,\alpha_{Ui}\left(\frac{Y_i}{U_i}\right)^{1/\sigma}. \tag{26}$$

The left-hand side is the effective cost, which is decreasing in buyers' intangible intensity $\bar{U}_i^b$. The right-hand side is the standard CES marginal revenue product. Solving for the optimal quantity of intangible services:

$$U_i^* = \left[\frac{\alpha_{Ui}\,p_i}{c_{Ui}}\right]^{\sigma} Y_i = \left[\frac{\alpha_{Ui}\,p_i\,(1+\theta\,\bar{U}_i^b)}{p_I}\right]^{\sigma} Y_i. \tag{27}$$

Because $\sigma > 0$, optimal intangible use is unambiguously increasing in $\bar{U}_i^b$. Thus, a firm whose buyers are more intangible-intensive faces a lower effective cost of intangibles and invests more. This is the core mechanism of the paper.

On the extensive margin, firm i adopts intangibles ($U_i > 0$) if and only if the profit gain exceeds the knowledge-adjusted fixed cost:

$$\pi_i(U_i^*) - \pi_i(0) \geq f \cdot (1-\kappa\,h_i). \tag{28}$$

The left-hand side is increasing in $\bar{U}_i^b$ (because lower cost implies larger gains from adoption) and the right-hand side is decreasing in h_i, so both the supply-chain and knowledge channels expand the set of adopters.

A stationary equilibrium consists of prices $\{p_i, w_i, p_I\}$, market tightness ξ, quantities $\{Y_i, L_i, M_i, U_i\}$, intangible adoption decisions $\{d_i\}$, and the knowledge distribution $\{h_i\}$ such that (i) each firm i solves Equation 25, choosing intermediate inputs and intangible purchases; (ii) the household maximizes utility subject to its budget constraint; (iii) wages satisfy Nash bargaining Equation 23

and vacancies satisfy free entry Equation 21; (iv) all markets clear:

$$Y_i = c_i + \sum_{j=1}^{N} x_{ji} + q_i^I \quad \text{for each } i, \tag{29}$$

$$\sum_{i=1}^{N} L_i = (1-u)\,\bar{L}, \tag{30}$$

$$\sum_{i=1}^{N} U_i = Q_I, \tag{31}$$

where q_i^I is firm i's output allocated to intangible services (positive only for $i \in \mathcal{I}$), and u is steady-state unemployment; and (v) the knowledge distribution $\{h_i\}$ is consistent with steady-state worker flows Equation 24.

6.6 Predictions

The model generates five predictions.

Proposition 1 (Supply-chain complementarity)**.** *Firm i's optimal intangible purchases U_i^* are increasing in its buyers' intangible intensity $\bar{U}_i^b$. Moreover, for a demand shock z_j hitting buyer j, $\partial^2 U_i^* / \partial z_j \, \partial \theta > 0$: the supplier's intangible purchase response to buyer demand shocks is amplified by the complementarity parameter θ.*

Proposition 2 (Extensive margin adoption)**.** *The probability that firm i adopts intangibles is increasing in $\bar{U}_i^b$ and in h_i. A new supply-chain link with a high-intangible buyer, or the hiring of a worker with intangible knowledge, can push a non-adopter above the adoption threshold.*

Proposition 3 (Knowledge transfer through hiring)**.** *When firm i hires a worker with $h = 1$, its effective fixed cost falls by $\kappa f / L_i$. For firms near the adoption threshold, this triggers a discrete increase in intangible purchases.*

Proposition 4 (Network amplification)**.** *The cost-reducing complementarity generates a network multiplier: an exogenous increase in one firm's intangible purchases reduces the effective cost for its suppliers, which raises their investment, which in turn benefits their own suppliers, and so on. The aggregate effect exceeds the partial equilibrium effect by a factor that depends on θ, σ, and the spectral properties of the sales-share matrix $\mathbf{S}$.*

Proposition 5 (General equilibrium dampening)**.** *The endogenous price p_I of intangible services partially offsets the complementarity-driven adoption wave. The dampening is stronger when the intangible-producing sector has steeper supply curves (lower ρ).*

Predictions 1–2 correspond to our demand shock decomposition (Figure 4 and Table 2) and B2B event study (Figure 5a). Prediction 3 corresponds to the hiring event study (Figure 5b). Predictions 4–5 are quantified in the counterfactual exercises below.

6.7 Calibration

We calibrate the model to the Dominican Republic economy. Table 4 summarizes all parameters, their values, and identification sources. Two structural parameters—θ and f—are calibrated directly in the general equilibrium. The complementarity parameter θ is identified by the semi-elasticity of supplier intangible purchases to demand shocks transmitted through intangible-purchasing buyers. That is, we regress the year-over-year change in the log of one plus intangible purchases on the two exposures z_{it}^H and z_{it}^L of Equation 6 and Equation 7, with the same sample, fixed effects, and controls as the local projection Equation 8, which yields $\hat{\beta}_H = 0.279$.[16] Log-linearizing Equation 27 at fixed prices and assuming that a unit demand shock raises an adopting buyer's intangible intensity one-for-one gives the mapping

$$\hat{\beta}_H = \frac{\sigma\,\theta\,\bar{\alpha}_U}{1+\theta\,\bar{U}^b} \quad \Longrightarrow \quad \theta = \frac{\hat{\beta}_H}{\sigma\,\bar{\alpha}_U - \hat{\beta}_H\,\bar{U}^b}, \tag{32}$$

where $\bar{\alpha}_U = 0.070$ is the mean intangible share among adopters in the 2019 calibration sample and $\bar{U}^b = 0.057$ is the mean buyer-weighted intangible intensity at baseline. This yields $\theta = 9.1$; ignoring the fixed-point term in the denominator would give 6.0.[17] The fixed cost f is set so that the equilibrium adoption rate matches the 52.3 percent of firms with positive intangible purchases in the 2019 calibration sample.

The third structural parameter, the knowledge-transfer κ, is identified by a partial-equilibrium match to the hiring event study. Given the data-implied distributions of h_i and firm employment L_i, we solve for the κ that produces a fraction of firms flipping from non-adopter to adopter (when a worker with $h = 1$ is added) equal to the extensive-margin coefficient at $k = 0$ from Figure 5b (4.1 pp). This yields $\kappa \approx 0.92$. For the general equilibrium counterfactuals we set $\kappa = 0$, which keeps the model baseline self-consistent with the data-based $\bar{U}^b$ construction described in Appendix A-IV; the $\kappa = 0$ choice is conservative in that a positive κ would amplify the aggregate effects reported below by adding a knowledge-transfer channel on top of the complementarity channel.[18]

The B2B event study of Figure 5a is not targeted in the calibration, which makes it available as an out-of-sample check on the model. We implement this check by simulating the new-link

[16] We target the investment semi-elasticity rather than the share response of Figure 4 because the model pins down the structural dependence of the investment level on buyer intangible intensity, whereas the intangible share is invariant to a proportional demand expansion (both intangibles and total purchases scale with output), so the share response does not identify θ. The share-margin local projection remains the reduced-form evidence for the mechanism; the level regression is the model-consistent analog.

[17] The same algebra delivers the model's asymmetric prediction: demand routed through non-adopting buyers cannot move their intangible intensity, so the mapping implies $\beta_L = 0$—the counterpart of the corollary to Proposition 1. The unit pass-through is a normalization rather than a model implication: in the model, buyer intensity responds through the extensive margin. Estimating the pass-through directly—regressing buyers' intangible-share changes on their own total propagated shock in the same specification—gives 0.61 percentage points per unit shock, well below the intensity a one-for-one response would imply. A smaller pass-through requires a larger θ to match the same $\hat{\beta}_H$, so $\theta = 9.1$, and the counterfactual magnitudes below, are lower bounds under this normalization.

[18] The calibration algorithm and computational details are described in Appendix A-IV.

event inside the calibrated model, holding prices and the fixed cost at their baseline values. Each non-adopting firm receives a first link to an intangible-intensive buyer with sales weight w, which raises its buyer-weighted intensity to $(1-w)\,\bar{U}_i^b + w\,\iota$, where ι is the new buyer's intangible intensity; matched controls receive a link of the same size to a buyer with $\iota = 0$, mirroring the empirical design. The model's predicted adoption response is the treated-minus-control share of non-adopters whose gain from adoption crosses the fixed cost. At the median sales weight of observed first links to intangible-industry buyers ($w = 0.09$), the predicted response is 1.1 percentage points when the new buyer has the average adopter's intangible intensity and 2.9 percentage points when it has the average intensity of the intangible-producing sector; at the mean weight ($w = 0.26$; the weight distribution is heavily right-skewed), the corresponding range is 2.8 to 7.2 percentage points. The untargeted empirical estimate of 4.0 percentage points falls inside this range.

Among the externally set parameters, the elasticity of substitution across intermediates η deserves comment, because it governs how the network responds in the counterfactuals. We set $\eta = 2$, under which intermediate expenditure shares respond to price changes: firms substitute toward cheaper suppliers, so that a firm whose intangible adoption lowers its costs captures a larger share of its buyers' intermediate spending. This generates endogenous network reweighting—the production network itself responds to changes in intangible purchases—rather than holding the observed linkages fixed. Micro estimates of this elasticity range from well below one in response to transitory shocks (Atalay, 2017; Boehm *et al.*, 2019) to about four in the long run (Peter and Ruane, 2026); our baseline $\eta = 2$ is an intermediate value within this range.

6.8 Counterfactual exercises

We use the calibrated model to quantify the aggregate importance of supply-chain complementarities in intangible inputs. Recall that we set $\kappa = 0$ in the baseline, so the counterfactuals below isolate the complementarity channel from the knowledge channel; an extension that calibrates $\kappa > 0$ using the hiring event study as a moment would amplify the aggregate numbers reported here.

Shutting down complementarities ($\theta \to 0$) Setting $\theta = 0$ eliminates the cost-reducing spillover, so each firm's effective intangible cost depends only on p_I, not on its buyers' intangible intensity. Table 5 reports the results. Aggregate intangible purchases fall by 15.2 percent and aggregate output declines by 4.10 percent. The adoption rate drops by 1.7 percentage points as marginal firms, whose gains from adoption depended on the cost reduction from intangible-intensive buyers, fall below the threshold. In partial equilibrium (holding p_I fixed), the decline in intangible purchases would be 26.2 percent; the endogenous price adjustment cushions about 42 percent of the impact, consistent with Proposition 5.

Table 4: Model parameters

Parameter	Description	Value	Source / identification
Directly from administrative data			
$\mathbf{\Omega}$	Input-output network	Firm-specific	VAT firm-to-firm transactions
$\alpha_{Li}, \alpha_{Mi}, \alpha_{Ui}$	Factor shares	Firm-specific	Cost shares from firm accounts
h_i	Knowledge share	Firm-specific	Employer-employee data: share of workers from $\mathcal{I}$
$\mathcal{I}$	Intangible producers	11% of firms	CIIU industry classification
Calibrated to match data moments			
θ	Complementarity	9.1	Match $\hat{\beta}_H = 0.279$ via Equation 32
f	Fixed cost of adoption	Calibrated	Match 52% adoption in calibration sample
κ	Knowledge transfer	0 (PE: 0.92)	Set to 0 in GE; PE match to hiring event study
From external literature			
σ	CES elasticity (inputs)	0.67	Lashkari *et al.* (2024)
η	CES elasticity (intermediates)	2.0	Intermediate value within literature
ε	Consumption elasticity	1.0	Unit elastic (Cobb-Douglas)
ρ	Intangible supply elasticity	3.0	See text
ν	Decreasing returns in $\mathcal{I}$	0.7	See text
r	Interest rate	0.05	Annual real rate
α_m	Matching elasticity	0.50	Petrongolo and Pissarides (2001)
ϕ	Bargaining weight	0.50	Hosios condition ($\phi = \alpha_m$)
δ_L	Separation rate	From data	TSS worker transitions
b	Flow value of unemployment	$0.4\bar{w}$	Standard calibration
μ	Matching efficiency	Calibrated	Match steady-state unemployment
c_v	Vacancy cost	Calibrated	Match average tightness

Table 5: Counterfactual results on the aggregate importance of intangible complementarities

	$\theta \to 0$ (GE) (1)	$\theta/2$ (GE) (2)	$\theta \to 0$ (PE*) (3)
Δ Aggregate intangible purchases	-15.2%	-6.1%	-26.2%
Δ Adoption rate	-1.7 pp	-0.1 pp	-4.8 pp
Δ Aggregate output	-4.10%	-3.57%	-4.67%
Δ Intangible price p_I	-21%	-9%	0%

Notes: The table reports the full general equilibrium counterfactuals using exact hat algebra. All endogenous variables (prices, output, wages, intangible purchases, adoption, and the intangible service price p_I) adjust simultaneously. The baseline ($\theta = 9.1$) is calibrated to match the investment semi-elasticity $\hat{\beta}_H = 0.279$ through the fixed-point mapping Equation 32. Column (1) shuts down the complementarity ($\theta \to 0$); column (2) halves it ($\theta/2$). Both recompute the equilibrium with endogenous p_I. Column (3) reports the partial equilibrium effect (holding p_I fixed). *Partial equilibrium: p_I held fixed at baseline level.

Robustness to the network elasticity The effects on intangible purchases are robust to the choice of η. With Cobb-Douglas intermediates ($\eta = 1$), where the network is held fixed, the complementarity-driven decline in intangibles is -16.7 percent, similar to the -15.2 percent under the baseline $\eta = 2$. The output effect is much smaller (-0.93 percent vs. -4.10 percent), because shutting down the network reweighting channel removes the reallocation of intermediate spending toward firms whose intangible adoption lowers their costs.

7 Conclusion

This paper shows that buyer-supplier relationships create complementarities in intangible inputs. Using administrative data covering the universe of firms in the Dominican Republic, we document that demand shocks originating from intangible-intensive buyers raise suppliers' own intangible purchases, while demand from other buyers does not. Forming a first link with an intangible-industry buyer triggers the supplier's own intangible adoption, and hiring a worker from an intangible-producing firm has a similar effect at the new employer. A full general equilibrium production network model—in which all prices, wages, output, and the intangible service price adjust simultaneously—implies that supply-chain complementarities account for about 15 percent of aggregate intangible purchases and 4.1 percent of output.

These findings have implications for policy and for the ongoing wave of AI adoption. Our results show that intangible input decisions are interdependent across firms. A supplier's incentive to invest depends on whether its buyers have invested. This interdependence creates the potential for both virtuous and vicious cycles. In a virtuous cycle, an anchor buyer's adoption of digital systems triggers supplier upgrades, which further reduce the buyer's costs and encourage deeper investment. In a vicious cycle, no firm invests because its partners have not, even when all would benefit from coordinated adoption. The fact that half of Dominican firms sit at zero intangible purchases, and more generally low intangible purchases in emerging markets, may partly reflect such coordination failures.

These findings suggest that programs aimed at promoting the adoption of intangible inputs—such as digitalization subsidies or technology adoption incentives—may have amplified effects when targeted at firms that are central in the production network. Our counterfactual analysis shows that eliminating the complementarity channel reduces aggregate intangible purchases by 15.2 percent and output by 4.1 percent, pointing to sizable multiplier effects that standard firm-level subsidies would miss. Identifying and supporting buyers whose adoption creates the largest downstream spillovers could catalyze adoption waves that propagate through the supply chain. This is especially relevant for the adoption of AI and related general-purpose technologies, whose value depends critically on compatibility across trading partners.

The knowledge transfer channel, which our counterfactuals switch off so that the aggregate numbers isolate the complementarity, points to a complementary policy lever. Facilitating worker mobility from intangible-producing sectors—through training programs, labor market intermedi-

ation, or reduced barriers to job switching—could lower the fixed cost of intangible adoption for firms that currently lack the internal expertise to implement new technologies. The combination of demand-side incentives (through anchor buyers) and supply-side measures (through knowledge diffusion) may be more effective than either alone.

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Online Appendix for:
"Intangible Input Complementarities Along Supply Chains"

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A-I Shock validation

We document that the network-propagated foreign demand shock used throughout the analysis—the supplier's total sales-weighted exposure to the Dhyne *et al.* (2021)-style export-demand shock of Equation 4, propagated through the lagged Leontief inverse—behaves as expected when regressed on standard firm-level activity. We estimate

$$Y_{it} = \beta z_{it}^{\text{total}} + \alpha_i + \mu_{jt} + \varepsilon_{it} \tag{33}$$

where Y_{it} denotes firm i's log sales and log VAT (business-to-business) sales, in levels and in first differences; $z_{it}^{\text{total}} = z_{it}^{H} + z_{it}^{L}$ is the demands shock; α_i are firm fixed effects; and μ_{jt} are industry $\times$ year fixed effects. We cluster standard errors clustered at the firm level.

The results of Table A1 suggest that the demand shock used in the analysis captures genuine variation in firm-level demand. A one-unit increase in the propagated foreign demand shock is associated with higher sales by about 10 percent and 15 percent in first differences. The firm's VAT sales—its total business-to-business sales, which constitute the production network we study—respond similarly, rising about 9 percent in levels and 6 percent in first differences, even though the latter is not statistically significant.

Table A1: Validation: firm-level responses to the network-propagated foreign demand shock

	ln sales (1)	ln VAT sales (2)	Δ ln sales (3)	Δ ln VAT sales (4)
z_{it}^{total}	0.095***	0.089**	0.144***	0.057
	(0.036)	(0.037)	(0.043)	(0.045)
Firm fixed effects	Yes	Yes	Yes	Yes
Industry × year fixed effects	Yes	Yes	Yes	Yes
R^2	0.905	0.899	0.267	0.245
Observations	266,747	266,747	207,761	207,761

Notes: The dependent variable is an indicator of firm-year activity. z_{it}^{total} denotes the sum of the high- and low-intangible-buyer exposures. Columns (1)–(2) use log levels of sales and VAT (business-to-business) sales; columns (3)–(4) use their first differences. All regressions include firm fixed effects and six-digit CIIU industry × year fixed effects. Standard errors clustered at the firm level are in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

A-II External validity in cross-country trade data

The complementarity we identify in Section 4 predicts that firms whose buyer base is more intangible-intensive should respond more to demand shocks routed through that base. While we do not have access to VAT transactions for multiple countries, we provide evidence of external validity of our findings by testing whether the same pattern is visible across countries. To do that, we map exporter countries to "suppliers," importer countries to "buyers," and bilateral trade flows from BACI to the network.

Each country c has a predetermined exposure to high-intangible destinations,

$$\bar{H}_c^B = \sum_{d \in \mathcal{H}} \omega_{cd}^{\text{pre}}, \tag{34}$$

where ω_{cd}^{pre} is the ratio of exports to high-intangible destinations over total exports, fixed at the 2013–2014 average computed from BACI HS6 bilateral trade. The cross-sectional test of the firm-level prediction is whether $\bar{H}_c^B$ predicts a country's subsequent change in intangible behavior:

$$\Delta y_c = a + b\, \bar{H}_c^B + \varepsilon_c, \tag{35}$$

where Δy_c denotes the change in the country's intangible measure between the 2023–2024 average and the 2013–2014 average.[19] b is positive under the complementarity reading.

We estimate two complementary specifications. The first specification combines BACI with the Global INTAN-Invest 2025 release, which reports the annual intangible-investment estimates for 32 advanced economies based on the Corrado *et al.* (2005, 2009) taxonomy. We define the high-intangible destinations as the set of countries whose 2013–2014 mean intangible-investment share of GDP exceeds the cross-country median. The outcome is the country's own change in that share between 2013–2014 and 2023–2024. Panel (a) of Figure A1 shows that countries whose exports were oriented toward intangible-intensive destinations in 2013–2014 (Luxembourg, Ireland, Estonia, Sweden, Finland, Denmark) experienced larger increases in their own intangible-investment share over the following decade than countries oriented toward less intangible-intensive destinations (Slovenia, Hungary, Czech Republic, Bulgaria). Estimating the underlying regression suggests that a ten-percentage-point increase in the predetermined share of exports going to intangible-intensive destinations is associated with a 1.3 percentage point larger increase in the country's intangible-investment share of GDP over the same window.[20]

To bring the cross-country evidence onto a broader sample, we estimate a second specification in which both the high-intangible destination classification and the outcome are constructed using BACI data.[21] In particular, we classify HS6 products as "intangible-content goods" using the

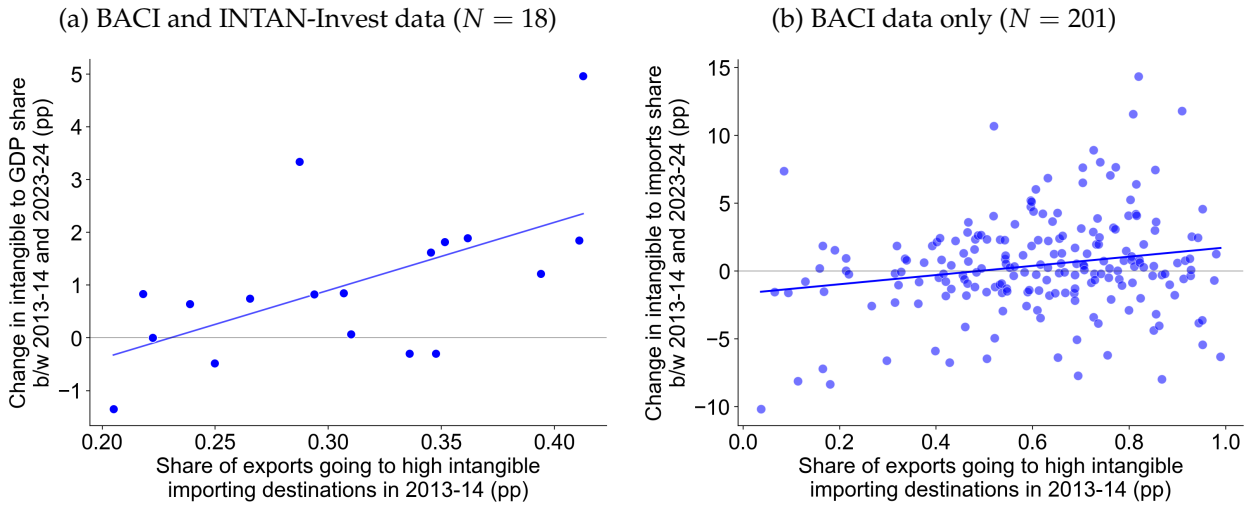
[19] Averaging over the two years reduces sensitivity to idiosyncratic annual fluctuations. The results are qualitatively the same if we use a single year.

[20] The association is similar in magnitude and significance using the earlier 2022–2023 endpoint (a 1.3-percentage-point larger increase per ten-percentage-point exposure) and is robust to excluding Ireland.

[21] One should note that BACI is goods trade only and therefore misses the services component of the Corrado *et al.*

standard Eurostat/OECD high-technology aggregate.[22] In this case, the set of high-intangible destinations are those in the top quartile of the predetermined intangible-import share. The cross-sectional outcome is the change in country c's own intangible-import share between 2013–2014 and 2023–2024.[23] As shown in panel (b) of Figure A1, the slope is positive. A regression implies a more modest relationship than in advanced economies, with a ten-percentage-point increase in the pre-determined share of exports going to intangible intensive destinations predicting a 0.34-percentage-point larger increase in the country's intangible-import share.

Figure A1: Export exposure to high-intangible destinations and intangible accumulation



Notes: In panel (a), markers denote advanced economies with INTAN-Invest data at both endpoints; the horizontal axis is the predetermined 2013–2014 share of each country's exports going to destinations whose intangible-investment share of GDP exceeds the cross-country median, and the vertical axis is the change in the country's own intangible-investment share of GDP between the 2013–2014 and 2023–2024 averages in percentage points. In panel (b), markers denote countries with BACI data at both endpoints; the horizontal axis is the top quartile of destinations by predetermined 2013–2014 mean intangible-import share, and the vertical axis is the change in each country's intangible-import share between the 2013–2014 and 2023–2024 averages in percentage points.

The two specifications offer suggestive evidence of external validation. Together they document that, across countries, the long-run accumulation of intangible activity is positively associated with the predetermined intangibility of trade-network partners. Two caveats apply. First, the design is purely cross-sectional, documenting a robust association rather than a causal effect. Second, the cross-country setting lacks the firm-level identifying ingredients that we exploit in

(2005, 2009) taxonomy, hence we view this classification as the closest goods-trade proxy for intangible flows.

[22] We classify intangible products if they fall into the following HS categories: chapter 30 (pharmaceuticals), chapter 88 (aircraft and aerospace), chapter 90 (scientific, optical, and medical instruments), and a set of ICT codes (8471, 8473, 8517, 8519, 8521, 8523, 8524, 8525, 8526, 8528, 8541, 8542). These are goods whose production is intensive in R&D, software, and design, and whose imports plausibly reflect downstream firms' demand for intangible-content inputs.

[23] The estimation sample is restricted to countries with at least 50 million US dollars of imports in 2013–2014 (filtering out micro-economies and tax havens whose import baskets are not informative) and at least 100 million US dollars of exports in 2013–2014 to retain a meaningful exporter, leaving 201 countries.

the main text. We therefore read Figure A1 as useful external corroboration that the same pattern is visible in macroeconomic data, while the identification of the mechanism remains a firm-level exercise.

A-III Model derivations

This appendix provides the full derivations of the model's key equations and the formal proofs of the five propositions stated in section 6.

A-III.1 Derivation of the first-order conditions

Intermediate input demand From the CES intermediate composite Equation 15, cost minimization implies that firm i's demand for input j is:

$$x_{ij} = \omega_{ij} \left(\frac{p_j}{P_i^M} \right)^{-\eta} M_i, \tag{36}$$

where $P_i^M = \left(\sum_j \omega_{ij} \, p_j^{1-\eta} \right)^{1/(1-\eta)}$ is the intermediate price index; throughout we normalize $\sum_j \omega_{ij} = 1$, which delivers the clean CES demand form above.

Intangible services demand The firm's profit function, conditional on adoption ($U_i > 0$), is:

$$\pi_i = p_i \, Y_i - w_i L_i - P_i^M M_i - c_{Ui} \, U_i,$$

where $c_{Ui} = p_I / (1 + \theta \bar{U}_i^b)$ is the effective price of intangible services from Equation 17 and Y_i is given by the CES production function Equation 14. The first-order condition for U_i is:

$$c_{Ui} = p_i \frac{\partial Y_i}{\partial U_i}.$$

From the CES production function:

$$\frac{\partial Y_i}{\partial U_i} = \alpha_{Ui} \left(\frac{Y_i}{U_i} \right)^{1/\sigma},$$

so that:

$$\frac{p_I}{1 + \theta \bar{U}_i^b} = p_i \, \alpha_{Ui} \left(\frac{Y_i}{U_i} \right)^{1/\sigma},$$

which is Equation 26. By analogous steps, the first-order conditions for labor and intermediates yield:

$$w_i = p_i \, \alpha_{Li} \left(\frac{Y_i}{L_i} \right)^{1/\sigma}, \tag{37}$$

$$P_i^M = p_i \, \alpha_{Mi} \left(\frac{Y_i}{M_i} \right)^{1/\sigma}. \tag{38}$$

Optimal intangible services (closed form) Solving Equation 26 for U_i:

$$U_i^{1/\sigma} = \frac{p_i\, \alpha_{Ui}\, (1 + \theta \bar{U}_i^b)}{p_I} \cdot Y_i^{1/\sigma}.$$

Raising both sides to the power σ:

$$U_i^* = \left[\frac{\alpha_{Ui}\, p_i\, (1 + \theta \bar{U}_i^b)}{p_I} \right]^{\sigma} Y_i. \tag{39}$$

Because $\sigma > 0$, the optimal quantity of intangible services U_i^* is unambiguously increasing in $\bar{U}_i^b$ for any value of σ. The cost-reducing formulation avoids the substitution-effect problem that arises with input-augmenting complementarity. That is, a lower effective cost always induces more investment, regardless of whether inputs are complements ($\sigma < 1$) or substitutes ($\sigma > 1$). This is the key advantage of modeling the complementarity as operating through the cost channel rather than through the production function directly.

A-III.2 Nash bargaining wage derivation

Let W_i denote the worker's value of employment at firm i and W_u the value of unemployment. The asset equations are:

$$rW_i = w_i + \delta_L (W_u - W_i), \tag{40}$$

$$rW_u = b + \lambda(\xi)(W_i - W_u), \tag{41}$$

where $\lambda(\xi) = \mu \xi^{1-\alpha_m}$ is the job-finding rate. The firm's value of a filled job satisfies:

$$rJ_i = p_i \cdot \text{MPL}_i - w_i - \delta_L J_i.$$

Nash bargaining maximizes $S_i = (W_i - W_u)^{\phi} J_i^{1-\phi}$, where ϕ is the worker's bargaining power. The first-order condition implies $\phi J_i = (1-\phi)(W_i - W_u)$, which, after substituting the asset equations and the free entry condition $J_i = c_v / q(\xi)$, yields:

$$w_i = (1-\phi) b + \phi \left(p_i \cdot \text{MPL}_i + c_v \xi \right),$$

which is Equation 23. Under the Hosios condition $\phi = \alpha_m$, the search externalities cancel and the equilibrium is constrained efficient.

A-III.3 Steady-state knowledge distribution

In steady state, the flow of workers into and out of each knowledge state must balance. Workers at intangible-producing firms ($j \in \mathcal{I}$) acquire $h = 1$ when they separate; all others carry $h = 0$. Separations occur at rate δ_L. The inflow of $h = 1$ workers into the unemployment pool is $\delta_L L_{\mathcal{I}}$ and

the outflow (through matching) is $\lambda(\xi)u\pi_h$, where π_h is the fraction of unemployed with $h = 1$. In steady state, the inflow equals the outflow:

$$\delta_L L_{\mathcal{I}} = \lambda(\xi)u\pi_h.$$

For the total pool, the steady-state condition $\delta_L \bar{L}(1-u) = \lambda(\xi)u$ implies:

$$\pi_h = \frac{L_{\mathcal{I}}}{\bar{L}(1-u)} \approx \frac{L_{\mathcal{I}}}{\bar{L}},$$

and under random matching $h_i = \pi_h$ for all firms. In the quantitative implementation, we replace this with firm-specific h_i computed directly from the employer-employee data.

A-III.4 Proofs of propositions

Proof of Proposition 1. Part (a): monotonicity in $\bar{U}_i^b$. From Equation 39, the optimal quantity of intangible services is:

$$U_i^* = \left[\frac{\alpha_{Ui}\, p_i\, (1+\theta\bar{U}_i^b)}{p_I}\right]^{\sigma} Y_i.$$

Since $\sigma > 0$, U_i^* is increasing in $(1+\theta\bar{U}_i^b)$ and hence in $\bar{U}_i^b$:

$$\frac{\partial U_i^*}{\partial \bar{U}_i^b} = \sigma\theta \cdot \frac{U_i^*}{1+\theta\bar{U}_i^b} > 0.$$

Because the complementarity enters as a price, this sign depends only on $\sigma > 0$, not on whether inputs are complements or substitutes. Two qualifications apply. First, this is a conditional comparative static: $\bar{U}_i^b$ itself responds in equilibrium, so the full response solves a fixed point; the derivative above is the direct effect, and the fixed point is well defined under the spectral-radius condition of Proposition 4. Second, it holds prices fixed; the general equilibrium feedback through p_I is the content of Proposition 5.

Part (b): the cross-partial $\partial^2 U_i^*/\partial z_j \partial\theta > 0$. The transmission from a buyer demand shock z_j to $\bar{U}_i^b$ operates through the *extensive* margin. Note first that the intensive margin alone cannot carry it: Equation 39 is homothetic in Y_j, so a pure scale shock raises U_j^* and Y_j proportionally and leaves the intensity U_j^*/Y_j—and hence $\bar{U}_i^b$—unchanged at fixed prices. The adoption margin breaks this invariance: the gain from adoption in Equation 28 scales with Y_j while the fixed cost $f(1-\kappa h_j)$ does not, so the probability that buyer j adopts, $d_j \in \{0,1\}$, is increasing in Y_j. Let $F_j(z_j) \equiv \partial \mathbb{E}[d_j]/\partial z_j \geq 0$, which is strictly positive when buyers have mass near the adoption threshold. Then

$$\frac{\partial \bar{U}_i^b}{\partial z_j} = s_{ij}\,\frac{U_j^*}{Y_j}\,F_j(z_j) > 0,$$

and combining with part (a),

$$\frac{\partial U_i^*}{\partial z_j} = \underbrace{\sigma\theta \, \frac{U_i^*}{1+\theta\bar{U}_i^b} \, s_{ij} \frac{U_j^*}{Y_j} F_j(z_j)}_{\text{complementarity channel}} + \underbrace{\frac{\partial U_i^*}{\partial Y_i} \frac{\partial Y_i}{\partial z_j}}_{\text{scale channel}}.$$

The scale channel is the ordinary demand effect, present for shocks through any buyer and independent of θ; the complementarity channel exists only if $\theta > 0$. Differentiating the first term with respect to θ yields a strictly positive expression at $\theta = 0$ and for all θ in the region where the spectral-radius condition holds, which establishes $\partial^2 U_i^* / \partial z_j \partial\theta > 0$.

Corollary (identification). The two channels separate exactly in the intangible *share*: the scale channel raises U_i^* and Y_i proportionally and cancels in U_i^*/Y_i, while the complementarity channel does not. The model therefore predicts a positive share response to demand routed through intangible-using (adopting) buyers and a zero share response to demand routed through non-adopting buyers, whose intensity cannot move: $\beta_H > 0$ and $\beta_L = 0$ in the notation of Equation 8. This is the model's rationalization of the empirical design in section 4. □ □

Proof of Proposition 2. Firm i adopts intangibles ($d_i = 1$) if and only if:

$$\Delta\pi_i \equiv \pi_i(U_i^*) - \pi_i(0) \geq f(1 - \kappa h_i).$$

Using the CES production function, the profit gain from investing U_i^* rather than zero is:

$$\Delta\pi_i = p_i Y_i(U_i^*) - p_i Y_i(0) - c_{Ui} U_i^*.$$

Substituting the optimal U_i^* from Equation 39, the gain $\Delta\pi_i$ is increasing in U_i^* (since the firm optimally chooses U_i^* to maximize this gain, and a higher U_i^* is induced by a lower cost c_{Ui}). Since $c_{Ui} = p_I(r+\delta)/(1+\theta\bar{U}_i^b)$ is decreasing in $\bar{U}_i^b$:

$$\frac{\partial \Delta\pi_i}{\partial \bar{U}_i^b} > 0.$$

Similarly, $\partial[f(1-\kappa h_i)]/\partial h_i = -f\kappa < 0$: a higher knowledge share lowers the adoption threshold. Both effects expand the set of adopters. □ □

Proof of Proposition 3. When firm i with L_i workers hires one worker with $h = 1$, the knowledge share changes by:

$$\Delta h_i = \frac{1}{L_i + 1} - \frac{h_i}{L_i + 1} \approx \frac{1 - h_i}{L_i}.$$

The effective fixed cost falls by:

$$\Delta f_i = -f\kappa\,\Delta h_i \approx -\frac{f\kappa(1-h_i)}{L_i}.$$

For a firm at the adoption margin ($\Delta\pi_i \approx f(1-\kappa h_i)$), this cost reduction can push it above the threshold, triggering a discrete jump from $U_i = 0$ to $U_i = U_i^* > 0$. Empirically, the hiring event study in Figure 5b identifies this extensive-margin flip as a $+4.1$ percentage point increase in the probability of any intangible purchase at $k = 0$. □ □

Proof of Proposition 4. From Equation 39, taking logs:

$$\ln U_i^* = \sigma \ln \alpha_{Ui} + \sigma \ln p_i - \sigma \ln p_I - \sigma \ln(r+\delta) + \sigma \ln(1+\theta \bar{U}_i^b) + \ln Y_i.$$

Log-linearizing around the equilibrium, with $\hat{U}_i = d \ln U_i$:

$$\hat{U}_i = \sigma \hat{p}_i + \hat{Y}_i - \sigma \hat{p}_I + \frac{\sigma\theta \bar{U}_i^b}{1+\theta \bar{U}_i^b}\, d \ln \bar{U}_i^b.$$

Since $\bar{U}_i^b = \sum_j s_{ij}(U_j/Y_j)$, we have $d \ln \bar{U}_i^b \approx \sum_j \tilde{s}_{ij}(\hat{U}_j - \hat{Y}_j)$, where $\tilde{s}_{ij}$ are expenditure-share-weighted contributions. Holding $\hat{p}_i$, $\hat{Y}_i$, and $\hat{p}_I$ fixed at their partial equilibrium values and stacking:

$$\hat{U} = \Phi\, \hat{z} + \operatorname{diag}\left(\frac{\sigma\theta \bar{U}^b}{1+\theta \bar{U}^b}\right) \mathbf{S}'\, \hat{U},$$

where $\Phi\hat{z}$ captures the direct effect of shocks. Solving:

$$\hat{U} = \left[\mathbf{I} - \operatorname{diag}\left(\frac{\sigma\theta \bar{U}^b}{1+\theta \bar{U}^b}\right) \mathbf{S}'\right]^{-1} \Phi\, \hat{z}.$$

The multiplier $\mathcal{M} = [\mathbf{I} - \operatorname{diag}(\cdot)\, \mathbf{S}']^{-1}$ is a Leontief-type inverse with positive entries when $\theta > 0$, because the cost-reducing complementarity implies positive feedback: higher buyer investment reduces supplier cost, inducing more supplier investment. The equilibrium exists and is unique when the spectral radius of $\operatorname{diag}(\sigma\theta \bar{U}^b/(1+\theta \bar{U}^b))\, \mathbf{S}'$ is less than 1. □ □

Proof of Proposition 5. In equilibrium, the intangible services market clears: $\sum_i U_i = Q_I$. The price of intangible services satisfies $p_I = MC_{\mathcal{I}}(Q_I)$, where $MC_{\mathcal{I}}$ is the marginal cost of the intangible-producing sector. When θ increases, aggregate demand for intangible services rises ($\sum_i I_i$ increases). With an upward-sloping supply curve ($\partial MC/\partial Q_I > 0$), p_I rises. From the FOC Equation 26, a higher p_I reduces optimal U_i^*, partially offsetting the complementarity effect. The dampening factor is $1/(1+\epsilon_D/\epsilon_S)$, where ϵ_D is the demand elasticity and $\epsilon_S = \rho$ is the supply elasticity of intangible services. □ □

A-IV Computational algorithm

The equilibrium is computed through a nested loop structure. This consists of an outer loop for the endogenous intangible price p_I, an inner loop for the intangible purchase profile $\{U_i\}$, and a bisection procedure for calibrating the fixed cost f. We describe each step in detail.

Step 1: Data inputs Load the I/O matrix $\mathbf{\Omega}$ (1,328,647 buyer-supplier pairs in 2019), firm characteristics (α_{Li}, α_{Mi}, α_{Ui}, revenue, employment) for 27,613 firms (the estimation sample of section 4 restricted to 2019 with B2B ratio ≥ 0.10), knowledge shares h_i (computed from employer-employee data), and intangible sector classification $\mathcal{I}$ based on CIIU codes.

Step 2: Externally set parameters Set $\sigma = 0.67$, $\eta = 2.0$, $r = 0.05$, $\alpha_m = 0.50$, $\phi = 0.50$, $\rho = 3.0$ from external sources (Table 4).

Step 3: Scale calibration From Equation 27, the optimal quantity of intangible services is $U_i^* = S \cdot \alpha_{Ui}^{\sigma} \cdot Y_i \cdot (1 + \theta \bar{U}_i^b)^{\sigma} \cdot p_I^{-\sigma}$, where S is a scale constant. We calibrate S once so that aggregate intangible purchases in the model match the data:

$$S = \frac{\sum_{i:d_i=1} U_i^{\text{data}}}{\sum_{i:d_i=1} \alpha_{Ui}^{\sigma} \cdot Y_i \cdot (1 + \theta \bar{U}_i^{b,\text{data}})^{\sigma} \cdot p_I^{-\sigma}},$$

where the sum is over firms that adopt intangibles in the data. This scale is held fixed across all counterfactuals.

Step 4: Inner loop (intangible purchases, given f and p_I)

1. Initialize U_i from observed intangible purchases.
2. Compute buyer-weighted intangible intensity: $\bar{U}_i^b = \sum_j s_{ij}(U_j/Y_j)$.
3. Compute optimal U_i^* from Equation 27.
4. Compute profit gains $\Delta\pi_i$ and adoption decisions: $d_i = \mathbf{1}\{\Delta\pi_i \geq f(1 - \kappa h_i)\}$.
5. Set $U_i = d_i \cdot U_i^*$.
6. Apply damped update: letting $\tilde{U}_i$ denote the value from step 5, set $U_i^{(k+1)} = 0.7\,\tilde{U}_i + 0.3\,U_i^{(k)}$ for stability.
7. Repeat steps 2–6 until $\max_i |U_i^{(k+1)} - U_i^{(k)}| < 10^{-6}$.

Step 5: Fixed cost calibration (bisection on f) The fixed cost f must be chosen so that the equilibrium adoption rate matches the data (52.3 percent of firms in the 2019 calibration sample adopt). Because the equilibrium adoption rate is a decreasing function of f (higher f means fewer firms find it profitable to adopt), we use bisection:

1. Set initial bounds $[f_{\text{lo}}, f_{\text{hi}}]$ from the gain distribution.
2. For $f_{\text{mid}} = (f_{\text{lo}} + f_{\text{hi}})/2$, solve the inner loop (Step 4).
3. If adoption rate $> 0.523 + 0.002$, set $f_{\text{lo}} = f_{\text{mid}}$; if $< 0.523 - 0.002$, set $f_{\text{hi}} = f_{\text{mid}}$.
4. Repeat until $|\text{adoption rate} - 0.523| < 0.002$.

This bisection is nested inside the price loop (Step 6).

Step 6: Outer loop (endogenous intangible price) The intangible service price p_I is determined by market clearing Equation 31. We compute p_I from the aggregate demand-supply balance:

$$p_I = \left(\frac{\sum_i U_i}{Q_{\mathcal{I}}^{\text{supply}}} \right)^{1/\rho},$$

where $Q_{\mathcal{I}}^{\text{supply}} = 0.5 \sum_{j \in \mathcal{I}} Y_j$ is the baseline supply capacity and $\rho = 3.0$. The outer loop iterates:

1. Given p_I, run Steps 4–5 to obtain equilibrium $\{U_i\}$.
2. Compute p_I^{new} from the formula above.
3. Update with damping: $p_I \leftarrow 0.5\, p_I^{\text{new}} + 0.5\, p_I$.
4. Repeat until $|p_I^{\text{new}} - p_I| / p_I < 10^{-4}$.

In practice, convergence is achieved in 6–10 outer iterations.

Step 7: Calibrate (θ, f, κ) The structural parameters are set sequentially:

- θ: set from the fixed-point mapping Equation 32, $\theta = \hat{\beta}_H / (\sigma \bar{\alpha}_U - \hat{\beta}_H \bar{U}^b)$, with $\hat{\beta}_H = 0.279$ (the coefficient on the high-intangible-buyer exposure z_{it}^H from a regression of the year-over-year change in the log of one plus intangible purchases on the exposures of Equation 6 and Equation 7; see subsection 6.7), $\bar{\alpha}_U = 0.070$ the mean intangible share among adopters in the 2019 calibration sample, and $\bar{U}^b = 0.057$ the mean baseline buyer-weighted intensity. This yields $\theta = 9.1$.
- f: calibrated via bisection (Step 5) to match the 50 percent non-adoption rate.

- κ: identified by a partial-equilibrium simulation of the hiring event study. For a candidate κ, we compute the share of non-adopting firms whose effective fixed cost $f(1-\kappa h_i)$ falls below the baseline adoption gain when a single worker with $h = 1$ is added (so $h_i \to \min(h_i + 1/L_i, 1)$). We bisect over κ until this simulated flip rate equals the extensive-margin hiring event-study coefficient of 4.1 pp, which yields $\kappa \approx 0.92$.

In the general equilibrium counterfactuals we set $\kappa = 0$ rather than using the calibrated value. This preserves self-consistency of the baseline in that the supplier-buyer sales-share matrix S and the baseline average buyer intangibles $\bar{U}^b$ are both constructed from data-observed intangible purchases, and with $\kappa > 0$ the model-implied adoption set diverges from the data set, breaking this consistency. Setting $\kappa = 0$ is therefore conservative, as the reported counterfactual numbers understate the aggregate effect of shutting down the complementarity channel, because a positive κ would add a reinforcing knowledge-transfer channel.

Step 8: Counterfactuals For each counterfactual ($\theta \to 0$, $\theta/2$ in GE, and $\theta \to 0$ in PE), we recompute Steps 4–6 holding f, S, and all other parameters at their baseline values. The key outputs are changes in aggregate intangible purchases, the adoption rate, and the equilibrium intangible price p_I.

Potential alpha for non-adopters In the data, non-adopting firms have $\alpha_{Ui} = 0$ by construction (they make no intangible purchases). In the model, α_{Ui} is a technology parameter representing the potential intangible intensity if the firm were to adopt. We assign non-adopters a potential α_{Ui} drawn from the empirical distribution of adopters' α_{Ui} (with replacement, using a fixed random seed for reproducibility). This ensures that the extensive margin responds to changes in θ, κ, and p_I. Some non-adopters have high potential α_{Ui} but face a fixed cost that exceeds their gain, while others have low potential and would not adopt even at zero fixed cost.

The random assignment means that the model's adoption set does not perfectly coincide with the data's, though the aggregate adoption rate matches by construction. We use the model-consistent adoption set as the baseline for counterfactual comparisons, ensuring internal consistency.



PUBLICATIONS

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